PHYSICS

ELECTROSTATICS_MEGA_XII



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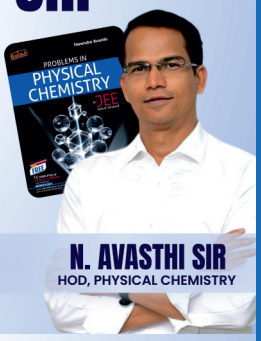

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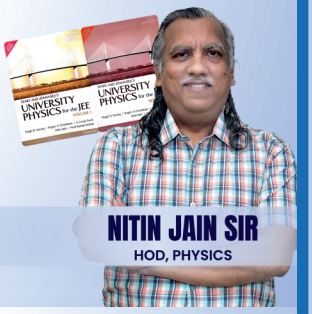


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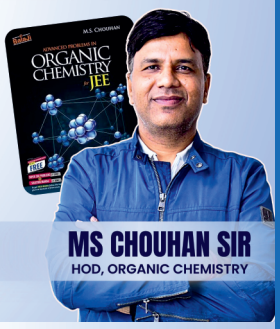


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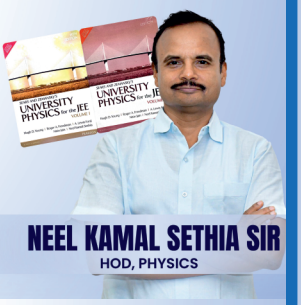


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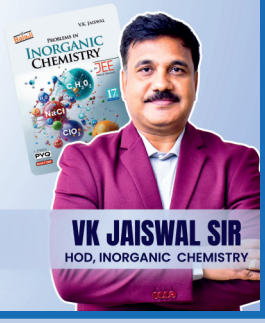


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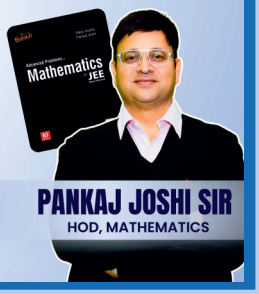


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PHYSICS

TARGET IIT JEE
MEGA-XII

ELECTROMAGNETIC INDUCTION
&
ALTERNATING CURRENT

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KEY CONCEPT

When a conductor is moved across a magnetic field, an electromotive force (emf) is produced in the conductor. If the conductors forms part of a closed circuit then the emf produced caused an electric current to flow round the circuit. Hence an emf (and thus a current) is induced in the conductor as a result of its movement across the magnetic field. This is known as **"ELECTROMAGNETIC INDUCTION."**

1. MAGNETIC FLUX :

$\phi = \vec{B} \cdot \vec{A} = BA\cos\theta$ weber for uniform $\vec{B}$.

$\phi = \int \vec{B} \cdot d\vec{A}$ for non uniform $\vec{B}$.

2. FARADAY'S LAWS OF ELECTROMAGNETIC INDUCTION :

(i) An induced emf is setup whenever the magnetic flux linking that circuit changes.

(ii) The magnitude of the induced emf in any circuit is proportional to the rate of change of the magnetic flux linking the circuit, $\varepsilon \alpha \frac{d\phi}{dt}$.

3. LENZ'S LAWS :

The direction of an induced emf is always such as to oppose the cause producing it .

4. LAW OF EMI :

$e = -\frac{d\phi}{dt}$. The negative sign indicated that the induced emf opposes the change of the flux .

5. EMF INDUCED IN A STRAIGHT CONDUCTOR IN UNIFORM MAGNETIC FIELD :

$$d\varepsilon = (\vec{V} \times \vec{B}) \cdot \vec{d\ell}$$

B = flux density in wb/m^2 ; $d\ell$ = length of the conductor (m) ;
V = velocity of the conductor (m/s) ;

6. COIL ROTATION IN MAGNETIC FIELD SUCH THAT AXIS OF ROTATION IS PERPENDICULAR TO THE MAGNETIC FIELD :

Instantaneous induced emf . $E = NAB\omega \sin\omega t = E_0 \sin\omega t$, where
N = number of turns in the coil ; A = area of one turn ; B = magnetic induction ;
ω = uniform angular velocity of the coil ; E_0 = maximum induced emf .

7. SELF INDUCTION & SELF INDUCTANCE :

When a current flowing through a coil is changed the flux linking with its own winding changes & due to the change in linking flux with the coil an emf is induced which is known as self induced emf & this phenomenon is known as self induction. This induced emf opposes the causes of Induction. The property of the coil or the circuit due to which it opposes any change of the current coil or the circuit is known as **SELF - INDUCTANCE**. It's unit is Henry.

Coefficient of Self inductance $L = \frac{\phi_s}{i}$ or $\phi_s = Li$

L depends only on ;

(i) shape of the loop &

(ii) medium

i = current in the circuit .

ϕ_s = magnetic flux linked with the circuit due to the current i .

self induced emf $e_s = \frac{d\phi_s}{dt} = -\frac{d}{dt}(Li) = -L\frac{di}{dt}$ (if L is constant)

8. MUTUAL INDUCTION :

If two electric circuits are such that the magnetic field due to a current in one is partly or wholly linked with the other, the two coils are said to be electromagnetically coupled circuits. Than any change of current in one produces a change of magnetic flux in the other & the latter opposes the change by inducing an emf within itself. This phenomenon is called **MUTUAL INDUCTION** & the induced emf in the latter circuit due to a change of current in the former is called **MUTUALLY INDUCED EMF**. The circuit in which the current is changed, is called the primary & the other circuit in which the emf is induced is called the secondary. The co-efficient of mutual induction (mutual inductance) between two electromagnetically coupled circuit is the magnetic flux linked with the secondary per unit current in the primary.

Mutual inductance = $M = \frac{\phi_m}{I_p} = \frac{\text{flux linked with secondary}}{\text{current in the primary}}$ mutually induced emf.

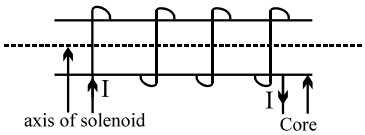
$E_m = \frac{d\phi_m}{dt} = -\frac{d}{dt}(MI) = -M\frac{dI}{dt}$ (If M is constant)

M depends on (1) geometery of loops (2) medium (3) orientation & distance of loops.

9. SOLENOID :

There is a uniform magnetic field along the axis the solenoid

(ideal : length >> diameter)



$B = \mu ni$ where ; μ = magnetic permeability of the core material ;

n = number of turns in the solenoid per unit length ; i = current in the solenoid ;

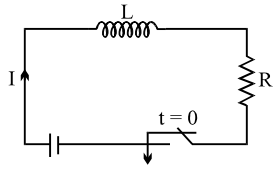
Self inductance of a solenoid $L = \mu_0 n^2 Al$; A = area of cross section of solenoid.

10. SUPER CONDUCTION LOOP IN MAGNETIC FIELD :

$R = 0$; $\varepsilon = 0$. Therefore ϕ_{total} = constant. Thus in a superconducting loop flux never changes. (or it opposes 100%)

11. ENERGY STORED IN AN INDUCTOR : $W = \frac{1}{2}LI^2$.

12. GROWTH OF A CURRENT IN AN L – R CIRCUIT :



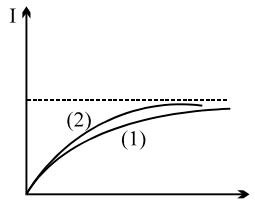
$I = \frac{E}{R}(1 - e^{-Rt/L})$. [If initial current = 0]

$\frac{L}{R}$ = time constant of the circuit .

$I_0 = \frac{E}{R}$.

(i) L behaves as open circuit at t = 0 [If $i = 0$]

(ii) L behaves as short circuit at $t = \infty$ always .

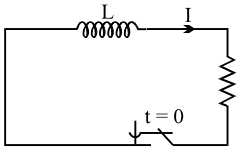


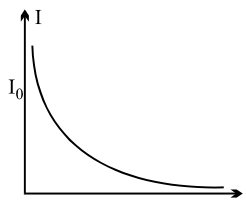
Curve (1) $\longrightarrow$ $\frac{L}{R}$ Large

Curve (2) $\longrightarrow$ $\frac{L}{R}$ Small

13. DECAY OF CURRENT :

Initial current through the inductor = I_0 ; Current at any instant $i = I_0 e^{-Rt/L}$

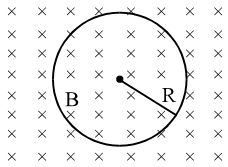




EXERCISE-I

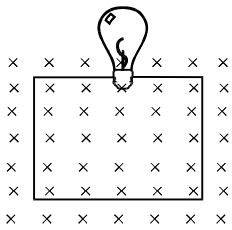
ONLY ONE OPTION IS CORRECT

1. a conducting loop of radius R is present in a uniform magnetic field B perpendicular the plane of the ring. If radius R varies as a function of time 't', as $R = R_0 + t$. The e.m.f induced in the loop is



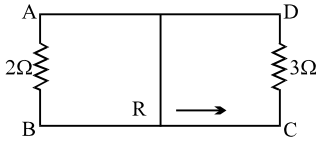
(A) $2\pi(R_0 + t)B$ clockwise (B) $\pi(R_0 + t)B$ clockwise
(C) $2\pi(R_0 + t)B$ anticlockwise (D) zero

2. A square wire loop of 10.0 cm side lies at right angles to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt. The bulb will shine with full brightness if Δt is equal to



(A) 20 ms (B) 0.02 ms (C) 2 ms (D) 0.2 ms

3. A rectangular loop with a sliding connector of length 10 cm is situated in uniform magnetic field perpendicular to plane of loop. The magnetic induction is 0.1 tesla and resistance of connector (R) is 1 ohm. The sides AB and CD have resistances 2 ohm and 3 ohm respectively. Find the current in the connector during its motion with constant velocity one metre/sec.

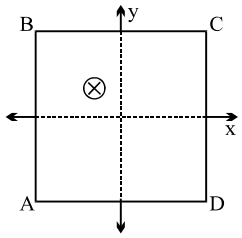


(A) $\frac{1}{110}$A (B) $\frac{1}{220}$A (C) $\frac{1}{55}$A (D) $\frac{1}{440}$A

4. The dimensions of permeability of free space can be given by
(A) $[MLT^{-2}A^{-2}]$ (B) $[MLA^{-2}]$ (C) $[ML^{-3}T^2A^2]$ (D) $[MLA^{-1}]$

5. A closed planar wire loop of area A and arbitrary shape is placed in a uniform magnetic field of magnitude B, with its plane perpendicular to magnetic field. The resistance of the wire loop is R. The loop is now turned upside down by 180° so that its plane again becomes perpendicular to the magnetic field. The total charge that must have flowed through the wire ring in the process is
(A) < AB/R (B) = AB/R (C) = 2AB/R (D) None

6. A square coil ABCD is placed in x-y plane with its centre at origin. A long straight wire, passing through origin, carries a current in negative z-direction. Current in this wire increases with time. The induced current in the coil is :



(A) clockwise (B) anticlockwise (C) zero (D) alternating

7. A vertical bar magnet is dropped from position on the axis of a fixed metallic coil as shown in fig-I. In fig-II the magnet is fixed and horizontal coil is dropped. The acceleration of the magnet & coil are a_1 & a_2 respectively then

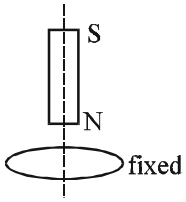


Fig-1

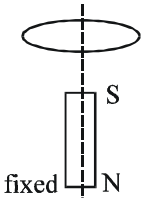
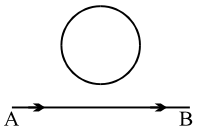


Fig-II

(A) $a_1 > g$, $a_2 > g$ (B) $a_1 > g$, $a_2 < g$ (C) $a_1 < g$, $a_2 < g$ (D) $a_1 < g$, $a_2 > g$

8. Two identical coaxial circular loops carry a current *i* each circulating in the same direction. If the loops approach each other
(A) the current in each will decrease
(B) the current in each will increase
(C) the current in each will remain the same
(D) the current in one will increase and in other will decrease

9. In the arrangement shown in given figure current from A to B is increasing in magnitude. Induced current in the loop will



(A) have clockwise direction (B) have anticlockwise direction
(C) be zero (D) oscillate between clockwise and anticlockwise

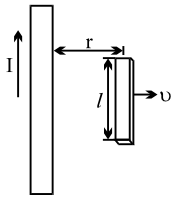
10. A thin wire of length 2m is perpendicular to the xy plane. It is moved with velocity $\vec{v} = (2\hat{i} + 3\hat{j} + \hat{k})\,m/s$ through a region of magnetic induction $\vec{B} = (\hat{i} + 2\hat{j})\ Wb/m^2$. Then potential difference induced between the ends of the wire :
(A) 2 volts (B) 4 volts (C) 0 volts (D) none of these

11. A long metal bar of 30 cm length is aligned along a north south line and moves eastward at a speed of 10 ms^{-1}. A uniform magnetic field of 4.0 T points vertically downwards. If the south end of the bar has a potential of 0 V, the induced potential at the north end of the bar is

(A) + 12 V (B) – 12 V

(C) 0 V (D) cannot be determined since there is not closed circuit

12. A conducting rod moves with constant velocity υ perpendicular to the long, straight wire carrying a current I as shown compute that the emf generated between the ends of the rod.



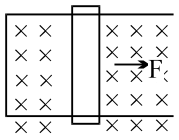
(A) $\frac{\mu_0 \upsilon I l}{\pi r}$ (B) $\frac{\mu_0 \upsilon I l}{2\pi r}$

(C) $\frac{2\mu_0 \upsilon I l}{\pi r}$ (D) $\frac{\mu_0 \upsilon I l}{4\pi r}$

13. A square loop of side a and resistance R is moved in the region of uniform magnetic field B (loop remaining completely insidefield) ,with a velocity v through a distance x. The work done is :

(A) $\frac{B\ell^2 vx}{R}$ (B) $\frac{2B^2\ell^2 vx}{R}$ (C) $\frac{4B^2\ell^2 vx}{R}$ (D) none

14. A rod closing the circuit shown in figure moves along a U shaped wire at a constant speed v under the action of the force F. The circuit is in a uniform magnetic field perpendicular to the plane. Calculate F if the rate of heat generation in the circuti is Q.



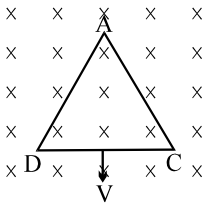
(A) F = Qv (B) F = $\frac{Q}{v}$

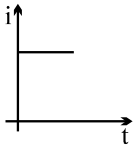
(C) F = $\frac{v}{Q}$ (D) F = $\sqrt{Qv}$

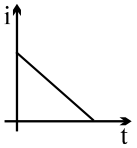
15. Two parallel long straight conductors lie on a smooth surface. Two other parallel conductors rest on them at right angles so as to form a square of side a initially. A uniform magnetic field B exists at right angles to the plane containing the conductors. They all start moving out with a constant velocity v. If r is the resistance per unit length of the wire the current in the circuit will be

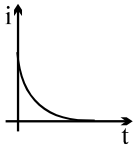
(A) $\frac{Bv}{r}$ (B) $\frac{Br}{v}$ (C) Bvr (D) Bv

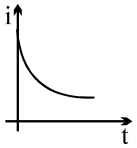
16. An equilateral triangular loop ADC of some finite magnetic field $\vec{B}$ as shown in the figure. At time t = 0, side DC of loop is at edge of the magnetic field. Magnetic field is perpendicular to the paper inwards (or perpendicular to the plane of the coil). The induced current versus time graph will be as



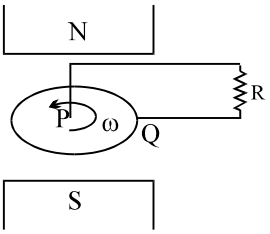
(A) 

(B) 

(C) 

(D) 

17. A metal disc rotates freely, between the poles of a magnet in the direction indicated. Brushes P and Q make contact with the edge of the disc and the metal axle.What current, if any, flows through R?



(A) a current from P to Q
(B) a current from Q to P
(C) no current, because the emf in the disc is opposed by the back emf
(D) no current, because the emf induced in one side of the disc is opposed by the emf induced in the other side.
(E) no current, because no radial emf is induced in the disc

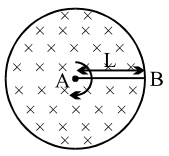
18. For L-R circuit, the time constant is equal to
(A) twice the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(B) ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(C) half the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(D) square of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

19. A rectangular coil of single turn, having area A, rotates in a uniform magnetic field B an angular velocity ω about an axis perpendicular to the field. If initially the plane of coil is perpendicular to the field, then the average induced e.m.f. when it has rotated through 90° is

(A) $\frac{\omega BA}{\pi}$ (B) $\frac{\omega BA}{2\pi}$ (C) $\frac{\omega BA}{4\pi}$ (D) $\frac{2\omega BA}{\pi}$

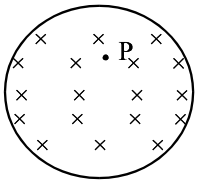
20. A ring of resistance 10Ω, radius 10 cm and 100 turns is rotated at a rate 100 revolutions per second about its diameter is perpendicular to a uniform magnetic field of induction 10 mT. The amplitude of the current in the loop will be nearly (Take : $\pi^2 = 10$)
(A) 200A (B) 2A (C) 0.002A (D) none of these

21. A copper rod AB of length L, pivoted at one end A, rotates at constant angular velocity ω, at right angles to a uniform magnetic field of induction B. The e.m.f developed between the mid point C of the rod and end B is



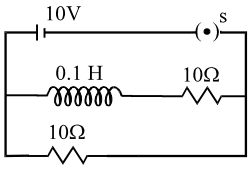
(A) $\frac{B\omega l^2}{4}$ (B) $\frac{B\omega l^2}{2}$ (C) $\frac{3B\omega l^2}{4}$ (D) $\frac{3B\omega l^2}{8}$

22. Figure shows a uniform magnetic field B confined to a cylindrical volume and is increasing at a constant rate. The instantaneous acceleration experienced by an electron placed at P is



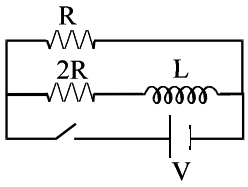
(A) zero (B) towards right (C) towards left (D) upwards

23. In the adjoining circuit, initially the switch S is open. The switch 'S' is closed at t = 0. The difference between the maximum and minimum current that can flow in the circuit is



(A) 2 Amp (B) 3 Amp (C) 1 Amp (D) nothing can be concluded

24. The ratio of time constant in build up and decay in the circuit shown in figure is

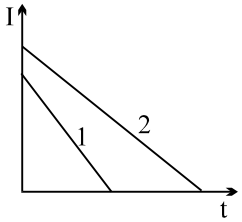


(A) 1 : 1 (B) 3 : 2 (C) 2 : 3 (D) 1 : 3

25. A curren of 2A is increasing at a rate of 4 A/s through a coil of inductance 2H. The energy stored in the inductor per unit time is

(A) 2 J/s (B) 1 J/s (C) 16 J/s (D) 4 J/s

26. Two identical inductance carry currents that vary with time according to linear laws (as shown in figure). In which of two inductance is the self induction emf greater?



(A) 1 (B) 2 (C) same (D) data are insufficient to decide

27. L, C and R represent physical quantities inductance, capacitance and resistance. The combination which has the dimensions of frequency is

(A) $\frac{1}{RC}$ and $\frac{R}{L}$ (B) $\frac{1}{\sqrt{RC}}$ and $\sqrt{\frac{R}{L}}$ (C) $\sqrt{LC}$ (D) $\frac{C}{L}$

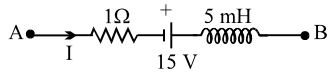
28. A coil of inductance 5H is joined to a cell of emf 6V through a resistance 10Ω at time t = 0. The emf across the coil at time t = $\ln \sqrt{2}$ s is :

(A) 3V (B) 1.5 V (C) 0.75 V (D) 4.5 V

29. A long solenoid of N turns has a self inductance L and area of cross section A. When a current i flows through the solenoid, the magnetic field inside it has magnitude B. The current i is equal to :

(A) BAN/L (B) BANL (C) BN/AL (D) B/ANL

30. The network shown in the figure is part of a complete circuit. If at a certain instant, the current I is 5A and it is decreasing at a rate of 10^3 As^{-1} then $V_B - V_A$ equals



(A) 20 V (B) 15 V (C) 10 V (D) 5 V

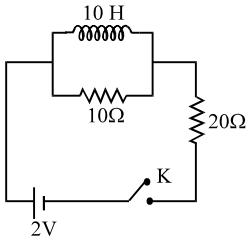
31. In **Problem 30**, if I is reversed in direction, then $V_B - V_A$ equals

(A) 5 V (B) 10 V (C) 15 V (D) 20 V

32. Two resistors of 10 Ω and 20 Ω and an ideal inductor of 10 H are connected to a 2 V battery as shown. The key K is inserted at time t = 0. The initial (t = 0) and final ($t \to \infty$) currents through battery are

(A) $\frac{1}{15}$ A, $\frac{1}{10}$ A (B) $\frac{1}{10}$ A, $\frac{1}{15}$ A

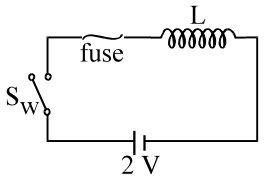
(C) $\frac{2}{15}$ A, $\frac{1}{10}$ A (D) $\frac{1}{15}$ A, $\frac{2}{25}$ A



33. A small coil of radius r is placed at the centre of a large coil of radius R, where R >> r. The coils are coplanar. The coefficient of mutual inductance between the coils is

(A) $\frac{\mu_0 \pi r}{2R}$ (B) $\frac{\mu_0 \pi r^2}{2R}$ (C) $\frac{\mu_0 \pi r^2}{2R^2}$ (D) $\frac{\mu_0 \pi r}{2R^2}$

34. In the circuit shown, the cell is ideal. The coil has an inductance of 4H and zero resistance. F is a fuse of zero resistance and will blow when the current through it reaches 5A. The switch is closed at t = 0. The fuse will blow :

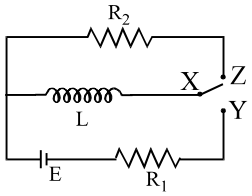


(A) just after t = 0 (B) after 2s (C) after 5s (D) after 10s

35. In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :

(A) $\frac{LE^2}{2R_1^2}$ (B) $\frac{LE^2}{2R_2^2}$

(C) $\frac{LE^2}{2R_1R_2}$ (D) $\frac{LE^2R_2}{2R_1^2}$



36. An induction coil stores 32 joules of magnetic energy and dissipates energy as heat at the rate of 320 watts when a current of 4 amperes is passed through it. Find the time constant of the circuit when the coil is joined across a battery.

(A) 0.2 s (B) 0.1 s (C) 0.3 s (D) 0.4 s

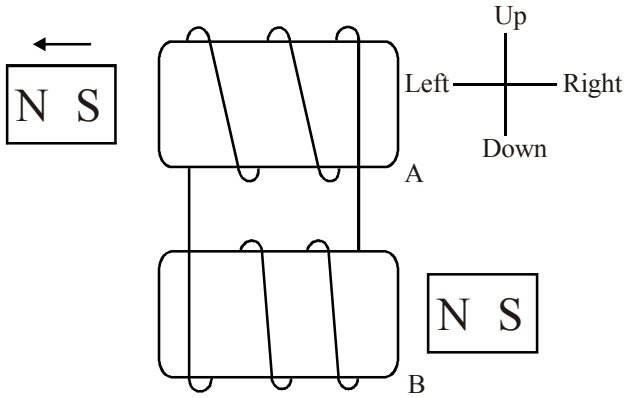
37. In a L-R decay circuit, the initial current at t = 0 is I. The total charge that has flown through the resistor till the energy in the inductor has reduced to one-fourth its initial value, is

(A) $\frac{LI}{R}$ (B) $\frac{LI}{2R}$ (C) $\frac{LI\sqrt{2}}{R}$ (D) None

38. A capacitor of capacitance 2 μF is charged to a potential difference of 12 V. It is then connected across an inductor of inductance 0.6 mH. The current in the circuit when the potential difference across the capacitor is 6 V is :

(A) 3.6 A (B) 2.4 A (C) 1.2 A (D) 0.6 A

39. A bar magnet was pulled away from a hollow coil A as shown. As the south pole came out of the coil, the bar magnet next to hollow coil B experienced a magnetic force



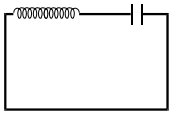
(A) to the right. (B) to the left. (C) upwards. (D) equal to zero.

40. The unit "henry" is equivalent to :

(A) volt-second/ampere (B) volt/second

(C) ampere-volt/second (D) ampere-second/volt

41. In an LC circuit, the capacitor has maximum charge q_0. The value of $\left.\frac{di}{dt}\right|_{max}$ is



(A) $\frac{q_0}{LC}$ (B) $\frac{q_0}{\sqrt{LC}}$

(C) $\frac{q_0}{LC}-1$ (D) none of these

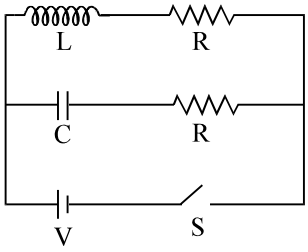
42. An inductor coil stores U energy when i current is passed through it and dissipates energy at the rate of P. The time constant of the circuit, when this coil is connected across a battery of zero internal resistance is

(A) $\frac{4U}{P}$ (B) $\frac{U}{P}$ (C) $\frac{2U}{P}$ (D) $\frac{2P}{U}$

43. A long straight wire is placed along the axis of a circular ring of radius R. The mutual inductance of this system is

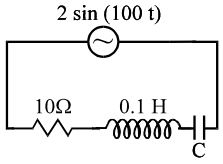
(A) $\frac{\mu_0 R}{2}$ (B) $\frac{\mu_0 \pi R}{2}$ (C) $\frac{\mu_0}{2}$ (D) 0

44. In the circuit shown in the figure, $R = \sqrt{\frac{L}{C}}$. Switch S is closed at time t = 0. The current through C and L would be equal after a time t equal to :



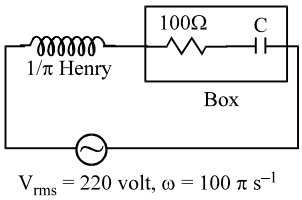
(A) CR (B) CR ln (2) (C) $\frac{L}{R\ln(2)}$ (D) LR

45. The power factor of the circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to



(A) 400 μF (B) 300 μF (C) 500 μF (D) 200 μF

46. In the circuit, as shown in the figure, if the value of R.M.S current is 2.2 ampere, the power factor of the box is



(A) $\frac{1}{\sqrt{2}}$ (B) 1 (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

47. When 100 V DC is applied across a solenoid a current of 1 A flows in it. When 100 V AC is applied across the same coil, the current drops to 0.5 A. If the frequency of the AC source is 50 Hz, the impedance and inductance of the solenoid are:

(A) 100Ω, 0.93 H (B) 200Ω, 1.0 H (C) 10Ω, 0.86H (D) 200Ω, 0.55 H

48. An inductive circuit contains resistance of 10 Ω and an inductance of 2.0 H. If an AC voltage of 120 V and frequency 60 Hz is applied to this circuit, the current would be nearly :

(A) 0.8 A (B) 0.48 A (C) 0.16 A (D) 0.32 A

49. The power in ac circuit is given by $P = E_{rms}I_{rms}\cos\phi$. The vale of cos ϕ in series LCR circuit at resonance is:

(A) zero (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

50. In ac circuit when ac ammeter is connected it reads i current if a student uses dc ammeter in place of ac ammeter the reading in the dc ammeter will be :

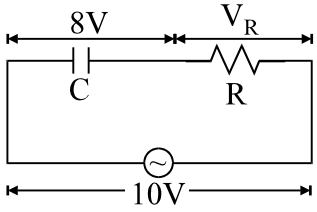
(A) $\frac{i}{\sqrt{2}}$ (B) $\sqrt{2}\,i$ (C) 0.637 i (D) zero

51. Let f = 50 Hz, and C = 100 μF in an AC circuit containing a capicator only. If the peak value of the current in the circuit is 1.57 A at t = 0. The expression for the instantaneous voltage across the capacitor will be

(A) E = 50 sin (100 πt – π/2) (B) E = 100 sin (50 πt)

(C) E = 50 sin 100 πt (D) E = 50 sin (100 πt + π/2)

52. In a series CR circuit shown in figure, the applied voltage is 10 V and the voltage across capacitor is found to be 8V. Then the voltage across R, and the phase difference between current and the applied voltage will respectively be

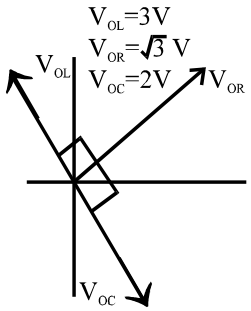


(A) 6V, $\tan^{-1}\left(\frac{4}{3}\right)$ (B) 3V, $\tan^{-1}\left(\frac{3}{4}\right)$ (C) 6V, $\tan^{-1}\left(\frac{5}{3}\right)$ (D) none

53. The phase difference between current and voltage in an AC circuit is π/4 radian. If the frequency of AC is 50 Hz, then the phase difference is equivalent to the time difference :

(A) 0.78 s (B) 15.7 ms (C) 0.25 s (D) 2.5 ms

54. The given figure represents the phasor diagram of a series LCR circuit connected to an ac source. At the instant t′ when the source voltage is given by V = $V_0\cos\omega t'$, the current in the circuit will be



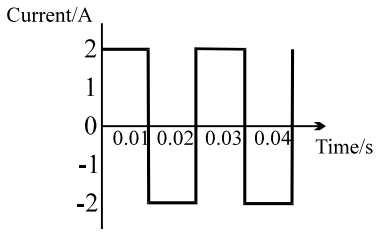
(A) I = $I_0\cos(\omega t' + \pi/6)$ (B) I = $I_0\cos(\omega t' - \pi/6)$

(C) I = $I_0\cos(\omega t' + \pi/3)$ (D) I = $I_0\cos(\omega t' - \pi/3)$

55. Power factor of an L-R series circuit is 0.6 and that of a C–R series circuit is 0.5. If the element (L, C, and R) of the two circuits are joined in series the power factor of this circuit is found to be 1. The ratio of the resistance in the L-R circuit to the resistance in the C–R circuit is

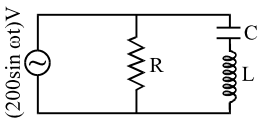
(An) $\frac{6}{5}$ (B) $\frac{5}{6}$ (C) $\frac{4}{3\sqrt{3}}$ (D) $\frac{3\sqrt{3}}{4}$

56. The direct current which would give the same heating effect in an equal constant resistance as the current shown in figure, i.e. the r.m.s. current, is



(A) zero (B) $\sqrt{2}$ A (C) 2A (D) $2\sqrt{2}$ A

57. In the circuit diagram shown, X_C = 100 Ω, X_L = 200 Ω and R = 100 Ω. The effective current through the source is

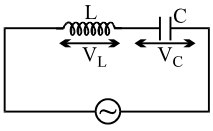


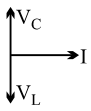
(A) 2 A (B) $\sqrt{2}$ A (C) 0.5 A (D) $2\sqrt{2}$ A

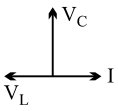
58. In series LR circuit X_L = 3R. Now a capacitor with X_C = R is added in series. Ratio of new to old power factor is

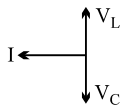
(A) 1 (B) 2 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

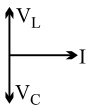
59. The current I, potential difference V_L across the inductor and potential difference V_C across the capacitor in circuit as shown in the figure are best represented vectorially as



(A) 

(B) 

(C) 

(D) 

60. A capacitor C = 2μF and an inductor with L = 10 H and coil resistance 5 Ω are in series in a circuit. When an alternating current of r.m.s. value 2A flows in the circuit, the average power in watts in the circuit is
(A) 100 (B) 50 (C) 20 (D) 10

61. A circuit has three elements, a resistance of 11Ω, a coil of inductive reactance 120Ω and a capacitive reactance of 120 Ω in series and connected to an A.C. source of 110 V, 60 Hz. Which of the three elements have minimum potential difference?
(A) Resistance (B) Capacitance
(C) Inductor (D) All will have equal potential difference

62. In a series R-L-C circuit, the frequency of the source is half of the resonance frequency. The nature of the circuit will be

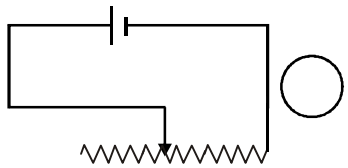
(A) capacitive (B) inductive (C) purely resistive (D) data insufficient

63. An a.c. source of voltage V and of frequency 50 Hz is connected to an inductor of 2H and negligible resistance. A current of r.m.s. value *I* flows in the coil. When the frequency of the voltage is changed to 400 Hz keeping the magnitude of V the same, the current is now

(A) 8*I* in phase with V (B) 4*I* and leading by 90° from V

(C) *I*/4 and lagging by 90° from V (D) *I*/8 and lagging by 90° from V

64. **Statement-1 :** When resistance of rheostat is increased, clockwise current is induced in the ring.
Statement-2 : Magnetic flux through the ring is out of the page and decreasing.



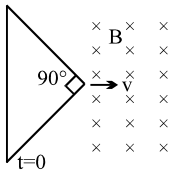
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

65. **Statement-1 :** The mutual inductance of two concentric conducting rings of different radii is maximum if the rings are also coplanar.
Statement-2 : For two coaxial conducting rings of different radii, the magnitude of magnetic flux in one ring due to current in other ring is maximum when both rings are coplanar.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

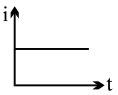
66. **Statement-1 :** Peak voltage across the resistance can be greater than the peak voltage of the source in an series LCR circuit.
Statement-2 : Peak voltage across the inductor can be greater than the peak voltage of the source in an series LCR circuit.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

67. **Statement-1 :** When an inductor coil is connected to a cell, initial current in it is zero.
Statement-2 : When an inductor coil is connected to a cell, the initial emf induced in it is zero.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

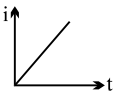
68. The figure shows an isosceles triangle wire frame with apex angle equal to $\pi/2$. The frame starts entering into the region of uniform magnetic field B with constant velocity v at t = 0. The longest side of the frame is perpendicular to the direction of velocity. If i is the instantaneous current through the frame then choose the alternative showing the correct variation of i with time.



(A) 

(B) 

(C) 

(D) 

69. A metallic rod of length L and mass M is moving under the action of two unequal forces F_1 and F_2 (directed opposite to each other) acting at its ends along its length. Ignore gravity and any external magnetic field. If specific charge of electrons is (e/m), then the potential difference between the ends of the rod is steady state must be

(A) $(F_1 - F_2)mL/eM$ (B) $2(F_1 - F_2)mL/eM$ (C) $[mL/eM]\, ln\, [F_1/F_2]$ (D) None

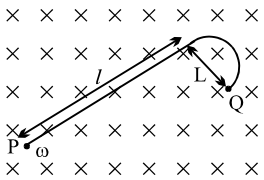
70. The magnetic field in a region is given by $\vec{B} = B_0\left(1+\frac{x}{a}\right)\hat{k}$. A square loop of edge - length d is placed with its edge along x & y axis. The loop is moved with constant velocity $\vec{V} = V_0\hat{i}$. The emf induced in the loop is

(A) $\frac{V_0B_0d^2}{a}$ (B) $\frac{V_0B_0d^2}{2a}$ (C) $\frac{V_0B_0a^2}{d}$ (D) None

71. The effective value of current i = 2 sin 100 π t + 2 sin(100 π t + 30°) is :

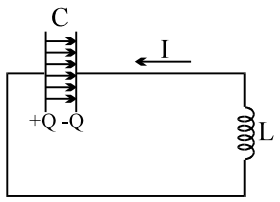
(A) $\sqrt{2}$ A (B) $2\sqrt{2+\sqrt{3}}$ (C) 4 (D) None

72. When a 'J' shaped conducting rod is rotating in its own plane with constant angular velocity w, about one of its end P, in a uniform magnetic field $\vec{B}$ directed normally into the plane of paper) then magnitude of emf induced across it will be



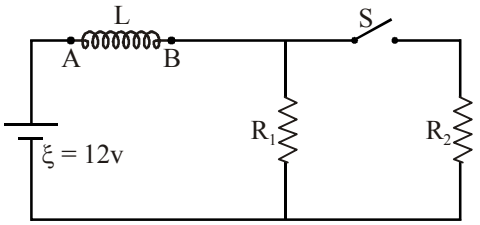
(A) $B\omega\sqrt{L^2+\ell^2}$ (B) $\frac{1}{2}B\omega L^2$ (C) $\frac{1}{2}B\omega(L^2+\ell^2)$ (D) $\frac{1}{2}B\omega\ell^2$

73. In the LC circuit, the current is in the direction shown and the charges on the capacitor plates have the signs shown. At this time



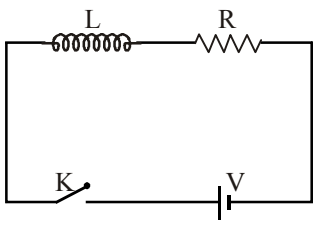
(A) I is increasing and Q is increasing (B) I is increasing and Q is decreasing
(C) I is decreasing and Q is increasing (D) I is decreasing and Q is decreasing

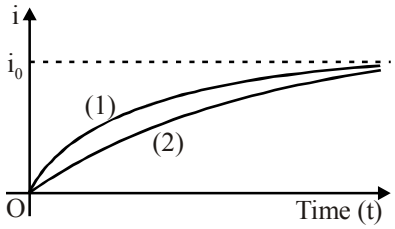
74. The circuit shown has been operating for a long time. The instant after the switch in the circuit labeled S is opened, what is the voltage across the inductor V_L and which labeled point (A or B) of the inductor is at a higher potential? Take $R_1 = 4.0\ \Omega$, $R_2 = 8.0\ \Omega$, and L = 2.5 H.



(A) V_L = 12 V ; Point A is at the higher potential (B) V_L = 12 V ; Point B is at the higher potential
(C) V_L = 6V ; Point A is at the higher potential (D) V_L = 6V ; Point B is at the higher potential

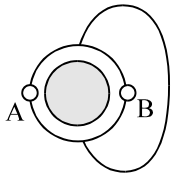
75. Refer to the circuit diagram and the corresponding graphs. The current rises when key K is pressed. With $R = R_1$ and $L = L_1$ the rise of current is shown by curve (1), while curve (2) shows the rise of current when $R = R_2$ and $L = L_2$. The maximum current is same for both curves, then :





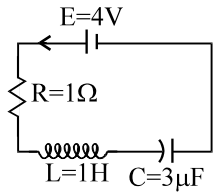
(A) $R_1 = R_2$, $L_1 > L_2$ (B) $R_1 > R_2$, $L_1 = L_2$ (C) $R_1 > R_2$, $L_1 < L_2$ (D) $R_1 = R_2$, $L_1 < L_2$

76. A very long solenoid perpendicular to the page generates a downward magnetic field whose magnitud increases with the time. A conducting wire loop around the solenoid contains two identical bulbs A and B which are glowing. Two points diametrically opposite on the wire loop are shorted with another wire lying to the right of bulb B in the plane of the page as shown in figure. After the shorting wire is inserted,



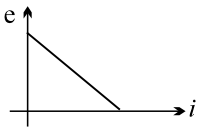
(A) Bulb A goes out and bulb B dims (B) Bulb A goes out and bulb B gets brighter
(C) Bulb B goes out and bulb A dims (D) Bulb B goes out and bulb A gets brighter

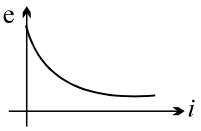
77. The current in the given circuit is increasing with a rate a = 4 amp/s. The charge on the capacitor at an instant when the current in the circuit is 2 amp will be :

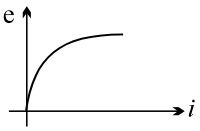


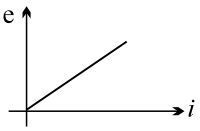
(A) 4 μC (B) 5 μC (C) 6 μC (D) none of these

78. In an L-R circuit connected to a battery of constant e.m.f. E switch S is closed at time t = 0. If e denotes the magnitude of induced e.m.f. across inductor and *i* the current in the circuite at any time t. Then which of the following graphs shows the variation of e with *i* ?

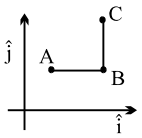
(A) 

(B) 

(C) 

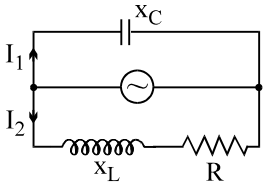
(D) 

79. There is a uniform magnetic field B normal to the xy plane. A conductor ABC has length AB = ℓ_1, parallel to the x-axis, and length BC = ℓ_2, parallel to the y-axis. ABC moves in the xy plane with velocity $v_x\hat{i} + v_y\hat{j}$. The potential difference between A and C is proportional to



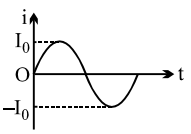
(A) $v_x\ell_1 + v_y\ell_2$ (B) $v_x\ell_2 + v_y\ell_1$ (C) $v_x\ell_2 - v_y\ell_1$ (D) $v_x\ell_1 - v_y\ell_2$

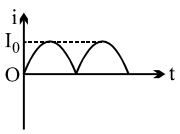
80. In the shown AC circuit phase different between currents I_1 and I_2 is

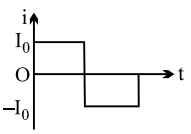


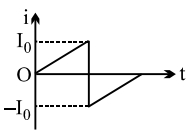
(A) $\frac{\pi}{2} - \tan^{-1}\frac{X_L}{R}$ (B) $\tan^{-1}\frac{X_L - X_C}{R}$ (C) $\frac{\pi}{2} + \tan^{-1}\frac{X_L}{R}$ (D) $\tan^{-1}\frac{X_L - X_C}{R} + \frac{\pi}{2}$

81. If I_1, I_2, I_3 and I_4 are the respective r.m.s. values of the time varying currents as shown in the four cases I, II, III and IV. Then identify the correct relations.



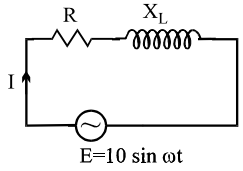






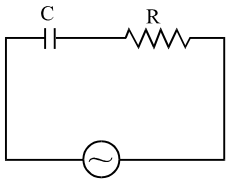
(A) $I_1 = I_2 = I_3 = I_4$ (B) $I_3 > I_1 = I_2 > I_4$ (C) $I_3 > I_4 > I_2 = I_1$ (D) $I_3 > I_2 > I_1 > I_4$

82. An ac-circuit having supply voltage E consists of a resistor of resistance 3Ω and an inductor of reactance 4Ω as shown in the figure. The voltage across the inductor at $t = \pi/\omega$ is



(A) 2 volts (B) 10 volts (C) zero (D) 4.8 volts

83. In the circuit shown if the emf of source at an instant is 5 V, the potential difference across capacitor at the same instant is 4 V. The potential difference across R at that instant may be



(A) 3V (B) 9V (C) $\frac{3}{\sqrt{2}}$V (D) none

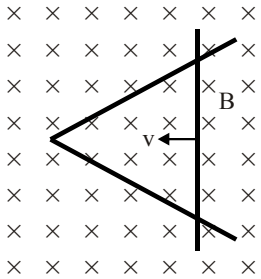
84. **Statement-1 :** When a circuit having large inductance is switched off sparking occurs at the switch.

Statement-2 : Emf induced in an inductor is given by $|\varepsilon| = L\left|\frac{di}{dt}\right|$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Question No. 85 to 87 (3 questions)

The mobile side of the triangular conducting frame made of uniform wire as shown in the figure is slid symmetrically at a uniform speed of v = 0.1 m/s along the two other sides. The horizontal frame is in a vertical homogeneous magnetic field with an induction of B = 0.4T. At a certain instant, the length of each side is ℓ = 1m and combined resistance of all three sides is 1Ω.



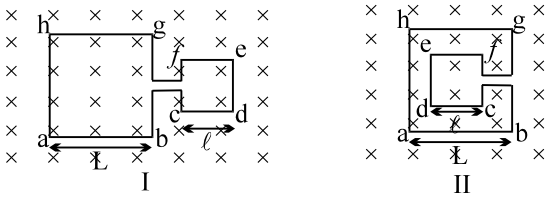
85. How much is the electromotive force induced in the circuit at the moment shown ?
(A) 0.04 V (B) 0.12 V (C) 0.06 V (D) None of these

86. How does the current change with time?
(A) it increases (B) it remains constant (C) it decreases (D) None of these

87. How much power is required for sliding at the instant shown if friction is negligible ?
(A) 8/15 mW (B) 1.6 mW (C) 4/3 mW (D) None of these

Question No. 88 to 91 (4 questions)

The adjoining figure shows two different arrangements in which two square wire frames of same resistance are placed in a uniform constantly decreasing magnetic field B.



88. The value of magnetic flux in each case is given by

(A) Case I : $\Phi = \pi(L^2 + \ell^2)B$; Case II : $\Phi = \pi(L^2 - \ell^2)B$

(B) Case I : $\Phi = \pi(L^2 + \ell^2)B$; Case II : $\Phi = \pi(L^2 + \ell^2)B$

(C) Case I : $\Phi = (L^2 + \ell^2)B$; Case II : $\Phi = (L^2 - \ell^2)B$

(D) Case I : $\Phi = (L + \ell)^2B$; Case II : $\Phi = \pi(L - \ell)^2B$

89. The direction of induced current in the case I is

(A) from a to b and from c to d (B) from a to b and from *f* to e

(C) from b to a and from d to c (D) from b to a and from e to *f*

90. The direction of induced current in the case II is

(A) from a to b and from c to d (B) from b to a and from *f* to e

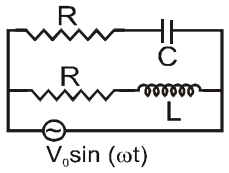
(C) from b to a and from c to d (D) from a to b and from d to c

91. If I_1 and I_2 are the magnitudes of induced current in the cases I and II, respectively, then

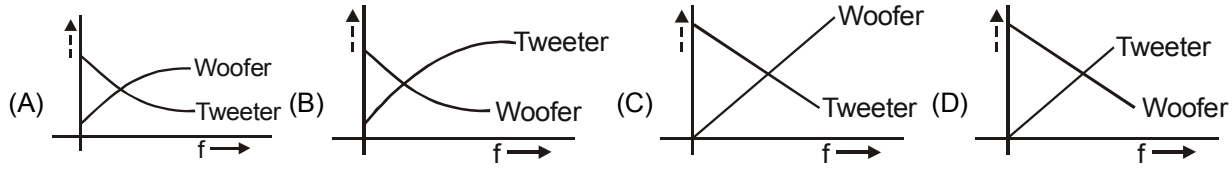
(A) $I_1 = I_2$ (B) $I_1 > I_2$ (C) $I_1 < I_2$ (D) nothing can be said

Question No. 92 to 94 (3 questions)

A loudspeaker system uses altemating current to amplify sound of certain frequencies. It consists of 2 speakers. Tweeter-which has smaller diameter produces high frequency sounds. Woofer-which has larger diameter produces low frequency sound. For purpose of circuit analysis, we can take both speakers to be of equal resistance R. The equivalent circuit is shown in the figure. The 2 speakers are connected to the amplifier via capacitance and inductance respectively. The capacitor in tweeter branch blocks the low frequency sound but passes the high frequency. The inductor in woofer branch does the opposite.



92. Which plot correctly represents rms current against frequency.



93. What is the frequency which is sounded equally loudly by both speakers

(A) $\frac{1}{2\pi}\sqrt{\frac{R^2}{L^2} - \frac{1}{LC}}$ (B) $\frac{1}{2\pi}\sqrt{\frac{4R^2}{L^2} - \frac{1}{LC}}$ (C) $\frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$ (D) $\frac{1}{2\pi\sqrt{LC}}$

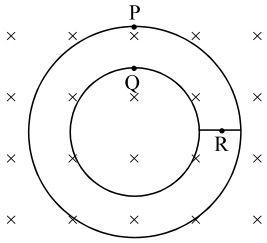
94. For a combination f L, R and C the current in woofer and tweeter are always found to have a phase difference of $\pi/2$. What is the relation between L, R and C.

(A) $L = 2R^2C$ (B) $L = \sqrt{2}\,R^2C$ (C) $L = R^2C$ (D) $L = \dfrac{R^2C}{\sqrt{2}}$

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

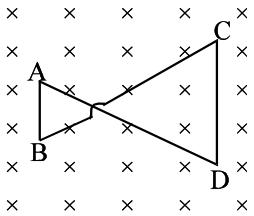
95. The dimension of the ratio of magnetic flux and the resistance is equal to that of :
(A) induced emf (B) charge (C) inductance (D) current

96. Figure shown plane figure made of a conductor located in a magnetic field along the inward normal to the plane of the figure. The magnetic field starts diminishing. Then the induced current



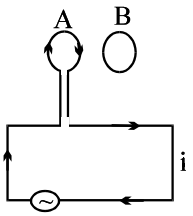
(A) at point P is clockwise (B) at point Q is anticlockwise
(C) at point Q is clockwise (D) at point R is zero

97. A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced currents in wires AB and CD are



(A) B to A and D to C (B) A to B and C to D
(C) A to B and D to C (D) B to A and C to D

98. Two circular coils A and B are facing each other as shown in figure. The current i through A can be altered



(A) there will be repulsion between A and B if i is increased
(B) there will be attraction between A and B if i is increased
(C) there will be neither attraction nor repulsion when i is changed
(D) attraction or repulsion between A and B depends on the direction of current. It does not depend whether the current is increased or decreased.

99. A bar magnet is moved along the axis of copper ring placed far away from the magnet. Looking from the side of the magnet, an anticlockwise current is found to be induced in the ring. Which of the following may be true?

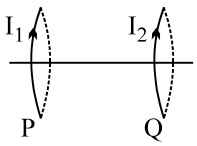
(A) The south pole faces the ring and the magnet moves towards it.

(B) The north pole faces the ring and the magnet moves towards it.

(C) The south pole faces the ring and the magnet moves away from it.

(D) The north pole faces the ring and the magnet moves away from it.

100. Two circular coils P & Q are fixed coaxially & carry currents I_1 and I_2 respectively



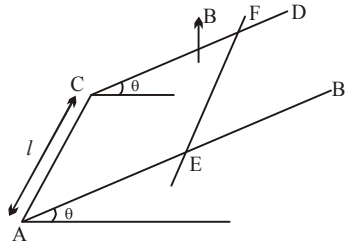
(A) if $I_2 = 0$ & P moves towards Q, a current in the samedirection as I_1 is induced in Q

(B) if $I_1 = 0$ & Q moves towards P, a current in the opposite direction to that of I_2 isinduced in P.

(C) when $I_1 \neq 0$ and $I_2 \neq 0$ are in the same direction then the two coils tend to move apart .

(D) when $I_1 \neq 0$ and $I_2 \neq 0$ are in opposite directions then the coils tends to move apart.

101. AB and CD are smooth parallel rails, separated by a distance *l*, and inclined to the horizontal at an angle θ. A uniform magnetic field of magnitude B, directed vertically upwards, exists in the region. EF is a conductor of mass m, carrying a current *i*. For EF to be in equilibrium,



(A) *i* must flow from E to F

(B) Bil = mg tan θ

(C) Bil = mg sin θ

(D) Bil = mg

102. In the **previous question**, if B is normal to the plane of the rails

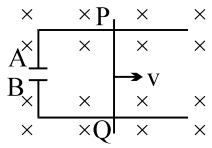
(A) Bil = mg tan θ

(B) Bil = mg sin θ

(C) Bil = mg cos θ

(D) equilibrium cannot be reached

103. A conducting rod PQ of length L = 1.0 m is moving with a uniform speed v = 20 m/s in a uniform magnetic field B = 4.0 T directed into the paper. A capacitor of capacity C = 10 μF is connected as shown in figure. Then



(A) $q_A = +800\ \mu C$ and $q_B = -800\ \mu C$

(B) $q_A = -800\ \mu C$ and $q_B = +800\ \mu C$

(C) $q_A = 0 = q_B$

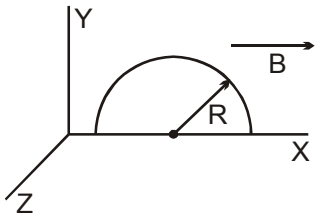
(D) charged stored in the capacitor increases exponentially with time

104. The e.m.f. induced in a coil of wire, which is rotating in a magnetic field, does not depend on

(A) the angular speed of rotation (B) the area of the coil

(C) the number of turns on the coil (D) the resistance of the coil

105. A semicircle conducting ring of radius R is placed in the xy plane, as shown in the figure. A uniform magnetic field is set up along the x-axis. No net emf, will be induced in the ring. if

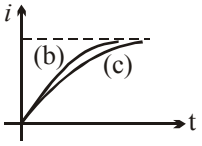


(A) it moves along the x-axis (B) it moves along the y-axis

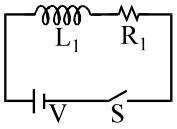
(C) it moves along the z-axis (D) it remains stationary

106. Current growth in two L-R circuits (b) and (c) as shown in figure (a). Let L_1, L_2, R_1 and R_2 be the corresponding values in two circuits. Then

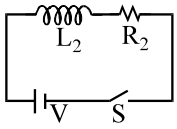
(a)



(b)

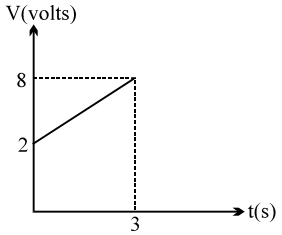


(c)



(A) $R_1 > R_2$ (B) $R_1 = R_2$ (C) $L_1 > L_2$ (D) $L_1 < L_2$

107. A circuit element is placed in a closed box. At time t = 0, constant current generator supplying a current of 1 amp, is connected across the box. Potential difference across the box varies according to graph shown in figure. The element in the box is :



(A) resistance of 2Ω (B) battery of emf 6V

(C) inductance of 2H (D) capacitance of 0.5F

108. The symbols L, C, R represent inductance, capacitance and resistance respectively. Dimension of frequency are given by the combination

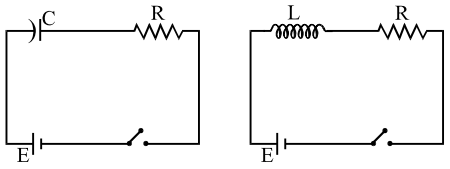
(A) $\frac{1}{RC}$ (B) $\frac{R}{L}$ (C) $\frac{1}{\sqrt{LC}}$ (D) $\frac{C}{L}$

109. An LR circuit with a battery is connected at t = 0. Which of the following quantities is not zero just after the circuit

(A) current in the circuit (B) magnetic field energy in the inductor

(C) power delivered by the battery (D) emf induced in the inductor

110. The switches in figures (a) and (b) are closed at t = 0

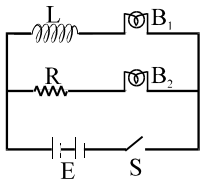


(a) (b)

(A) The charge on C just after t = 0 is EC. (B) The charge on C long after t = 0 is EC.
(C) The current in L just after t = 0 is E/R. (D) The current in L long after t = 0 is E/R.

111. For L – R circuit, the time constant is equal to
(A) twice the ratio of the energy stored in the magnetic field to the rate of the dissipation of energy in the resistance.
(B) the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance.
(C) half of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance.
(D) square of the ratio of the energy stored in the magnetic field to the rate of dissipation energy in the resistance.

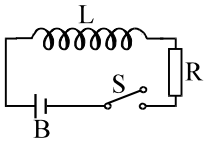
112. An inductor L, a resistance R and two identical bulbs B_1 and B_2 are connected to a battery through a switch S as shown in the figure. The resistance of coil having inductance L is also R. Which of the following statement gives the correct description of the happenings when the switch S is closed?



(A) The bulb B_2 lights up earlier than B_1 and finally both the bulbs shine equally bright.
(B) B_1 light up earlier and finally both the bulbs acquire equal brightness.
(C) B_2 lights up earlier and finally B_1 shines brighter than B_2.
(D) B_1 and B_2 light up together with equal brightness all the time.

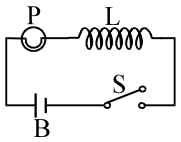
113. Which of the following quantities can be written in SI units in $Kgm^2A^{-2}s^{-3}$?
(A) Resistance (B) Inductance (C) Capacitance (D) Magnetic flux

114. In figure, the switch S is closed so that a current flows in the iron-core inductor which has inductance L and the resistance R. When the switch is opened, a spark is obtained in it at the contacts. The spark is due to



(A) a slow flux change in L (B) a sudden increase in the emf of the battery B
(C) a rapid flux change in L (D) a rapid flux change in R

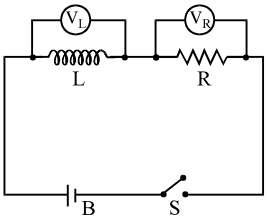
115. In figure, a lamp P is in series with an iron-core inductor L. When the switch S is closed, the brightness of the lamp rises relatively slowly to its full brightness than it would do without the inductor. This is due to



(A) the low resistance of P (B) the induced-emf in L
(C) the low resistance of L (D) the high voltage of the battery B

116. Two coil A and B have coefficient of mutual inductance M = 2H. The magnetic flux passing through coil A changes by 4 Weber in 10 seconds due to the change in current in B. Then
(A) change in current in B in this time interval is 0.5 A
(B) the change in current in B in this time interval is 2A
(C) the change in current in B in this time interval is 8A
(D) a change in current of 1A in coil A will produce a change in flux passing through B by 4 Weber.

117. An inductance L, resistance R, battery B and switch S are connected in series. Voltmeters V_L and V_R are connected across L and R respectively. When switch is closed :



(A) The initial reading in V_L will be greater than in V_R.
(B) The initial reading in V_L will be lesser than V_R.
(C) The initial readings in V_L and V_R will be the same.
(D) The reading in V_L will be decreasing as time increases.

118. Two different coils have self inductance 8 mH and 2 mH. The current in one coil is increased at a constant rate. The current in the second coild is also increased at the same constant rate. At a certain instant of time, the power given to the two coils is the same. At that time the current, the induced voltage and the energy stored in the first coil are I_1, V_1 and W_1 respectively. Corresponding values for the second coil at the same instant are I_2, V_2 and W_2 respectively. Then :

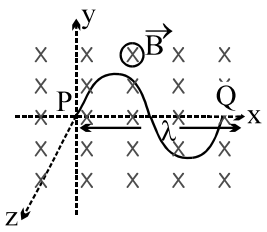
(A) $\frac{I_1}{I_2} = \frac{1}{4}$ (B) $\frac{I_1}{I_2} = 4$ (C) $\frac{W_2}{W_1} = 4$ (D) $\frac{V_2}{V_1} = \frac{1}{4}$

119. Resonance occurs in a series L-C-R circuit when the frequency of the applied emf is 1000 Hz. Then :
(A) when f = 900 Hz, the circuit behaves as a capacitative circuit
(B) the impedance of the circuit is maximum at f = 1000 Hz
(C) at resonance the voltage accross L and voltage accross C differ in phase by 180°
(D) if the value of C is doubled resonance occurs at f = 2000 Hz

EXERCISE-II

1. The horizontal component of the earth's magnetic field at a place is 3×10^{-4} T and the dip is $\tan^{-1}(4/3)$. A metal rod of length 0.25 m placed in the north-south position is moved at a constant speed of 10cm/s towards the east. Find the e.m.f. induced in the rod.

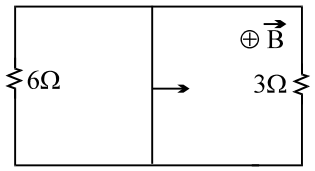
2. A wire forming one cycle of sine curve is moved in x-y plane with velocity $\vec{V} = V_x\hat{i} + V_y\hat{j}$. There exist a magnetic field $\vec{B} = -B_0\hat{k}$. Find the motional emf develop across the ends PQ of wire.



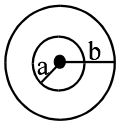
3. A conducting circular loop is placed in a uniform magnetic field of 0.02 T, with its plane perpendicular to the field . If the radius of the loop starts shrinking at a constant rate of 1.0 mm/s, then find the emf induced in the loop, at the instant when the radius is 4 cm.

4. Find the dimension of the quantity $\frac{L}{RCV}$, where symbols have usual meaining.

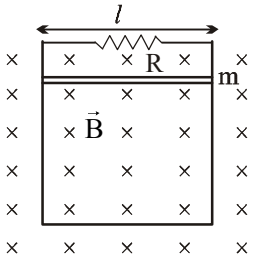
5. A rectangular loop with a sliding connector of length *l* = 1.0 m is situated in a uniform magnetic field B = 2T perpendicular to the plane of loop. Resistance of connector is r = 2Ω. Two resistances of 6Ω and 3Ω are connected as shown in figure. Find the external force required to keep the connector moving with a constant velocity v = 2 m/s.



6. Two concentric and coplanar circular coils have radii a and b(>>a)as shown in figure. Resistance of the inner coil is R. Current in the outer coil is increased from 0 to i , then find the total charge circulating the inner coil.

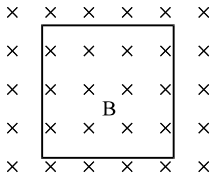


7. A horizontal wire is free to slide on the long vertical rails of a conducting frame as shown in figure. The wire has a mass m and length *l* and the resistance of the circuit is R. If a uniform magnetic field B is directed perpendicular to the frame,then find the terminal speed of the wire as it falls remaining in contact with rails under the force of gravity.

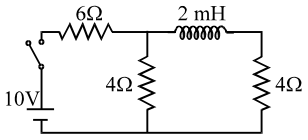


8. A metal rod of resistance 20Ω is fixed along a diameter of a conducting ring of radius 0.1 m and lies on x-y plane. There is a magnetic field $\vec{B}$ = (50T)$\hat{k}$. The ring rotates with an angular velocity ω = 20 rad/sec about its axis. An external resistance of 10Ω is connected across the centre of the ring and rim. Find the current through external resistance.

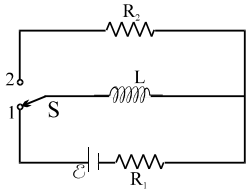
9. A uniform magnetic field of 0.08 T is directed into the plane of the page and perpendicular to it as shown in the figure. A wire loop in the plane of the page has constant area 0.010 m^2. The magnitude of magnetic field decrease at a constant rate of 3.0×10^{-4} Ts^{-1}. Find the magnitude and direction of the induced emf in the loop.



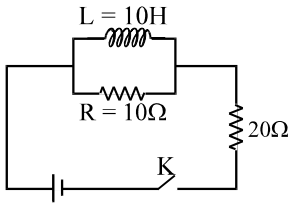
10. In the given current, find the ratio of i_1 to i_2 where i_1 is the initial (at t = 0) current and i_2 is steady state (at t = ∞) current through the battery.



11. In the circuit shown, initially the switch is in position 1 for a long time. Then the switch is shifted to position 2 for a long time. Find the total heat produced in R_2.



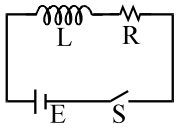
12. Two resistors of 10Ω and 20Ω and an ideal inductor of 10H are connected to a 2V battery as shown. The key K is shorted at time t = 0. Find the initial (t = 0) and final ($t \to \infty$) currents through battery.



13. There exists a uniform cylindrically symmetric magnetic field directed along the axis of a cylinder but varying with time as B = kt. If an electron is released from rest in this field at a distance of 'r' from the axis of cylinder, its acceleration, just after it is released would be (e and m are the electronic charge and mass respectively)

14. An emf of 15 volt is applied in a circuit containing 5 H inductance and 10 Ω resistance. Find the ratio of the currents at time $t = \infty$ and t = 1 second.

15. In the circuit shown in figure switch S is closed at time t = 0. Find the charge which passes through the battery in one time constant.

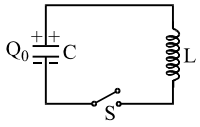


16. Two coils, 1 and 2, have a mutual inductance = M and resistances R each. A current flows in coil 1, which varies with time as: $I_1 = kt^2$, where K is a constant and 't' is time. Find the total charge that has flown through coil 2, between t = 0 and t = T.

17. In a L–R decay circuit, the initial current at t = 0 is I. Find the total charge that has flown through the resistor till the energy in the inductor has reduced to one–fourth its initial value.

18. An inductor of inductance 2.0 mH,is connected across a charged capacitor of capacitance 5.0 μF, and the resulting LC circuit is set oscillating at its natural frequency. Let Q denote the instantaneous charge on the capacitor, and I the current in the circuit. It is found that the maximum value of Q is 200 μC.
(a) when Q = 100μC, what is the value of |dI / dt|?
(b) when Q = 200 μC ,what is the value of I ?
(c) Find the maximum value of I.
(d) when I is equal to one half its maximum value, what is the value of |Q|

19. A capacitor C with a charge Q_0 is connected across an inductor through a switch S. If at t = 0, the switch is closed, then find the instantaneous charge q on the upper plate of capacitor.



20. A coil of resistance 300Ω and inductance 1.0 henry is connected across an alternating voltage of frequency 300/2π Hz. Calculate the phase difference between the voltage and current in the circuit.

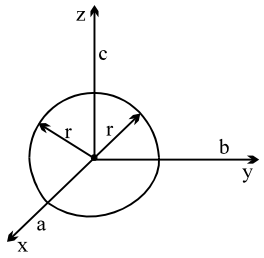
21. Find the value of an inductance which should be connected in series with a capacitor of 5 μF, a resistance of 10Ω and an ac source of 50 Hz so that the power factor of the circuit is unity.

22. In an L-R series A.C circuit the potential difference across an inductance and resistance joined in series are respectively 12 V and 16 V. Find the total potential difference across the circuit.

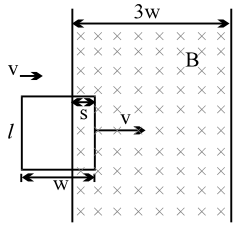
23. When 100 volt D.C. is applied across a coil, a current of one ampere flows through it, when 100 V ac of 50 Hz is applied to the same coil, only 0.5 amp flows. Calculate the resistance and inductance of the coil.

24. A 50 W, 100 V lamp is to be connected to an ac mains of 200 V, 50 Hz. What capacitance is essential to be put in series with the lamp.

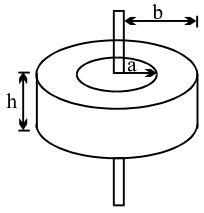
25. A wire is bent into 3 circular segments of radius r = 10 cm as shown in figure.
Each segment is a quadrant of a circle, ab lying in the xy plane, bc lying in the yz plane & ca lying in the zx plane.
(i) if a magnetic field B points in the positive x direction, what is the magnitude of the emf developed in the wire, when B increases at the rate of 3 mT/s ?
(ii) what is the direction of the current in the segment bc.



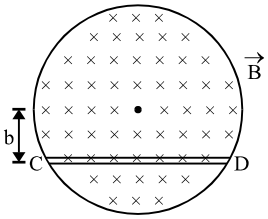
26. A rectangular loop of dimensions *l* & w and resistance R moves with constant velocity V to the right as shown in the figure. It continues to move with same speed through a region containing a uniform magnetic field B directed into the plane of the paper & extending a distance 3 W. Sketch the flux, induced emf & external force acting on the loop as a function of the distance.



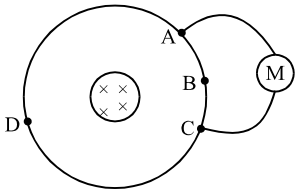
27. A long straight wire is arranged along the symmetry axis of a toroidal coil of rectangular cross-section, whose dimensions are given in the figure. The number of turns on the coil is N, and relative permeability of the surrounding medium is unity. Find the amplitude of the emf induced in this coil, if the current $i = i_m \cos \omega t$ flows along the straight wire.



28. A uniform magnetic field $\vec{B}$ fills a cylindrical volumes of radius R. A metal rod CD of length *l* is placed inside the cylinder along a chord of the circular cross-section as shown in the figure. If the magnitude of magnetic field increases in the direction of field at a constant rate dB/dt, find the magnitude and direction of the EMF induced in the rod.



29. A variable magnetic field creates a constant emf E in a conductor ABCDA. The resistances of portion ABC, CDA and AMC are R_1, R_2 and R_3 respectively. What current will be shown by meter M? The magnetic field is concentrated near the axis of the circular conductor.

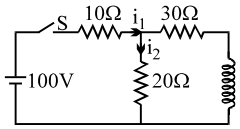


30. A charged ring of mass m = 50 gm, charge 2 coulomb and radius R = 2m is placed on a smooth horizontal surface. A magnetic field varying with time at a rate of (0.2 t) Tesla/sec is applied on to the ring in a direction normal to the surface of ring. Find the angular speed attained in a time t_1 = 10 sec.

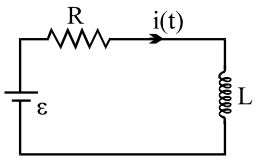
31. Find the values of i_1 and i_2

(i) immediately after the switch S is closed. (ii) long time later, with S closed.

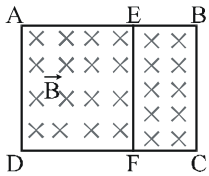
(iii) immediately after S is open. (iv) long time after S is opened.



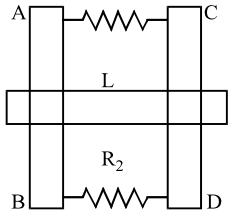
32. Suppose the emf of the battery, the circuit shown varies with time t so the current is given by $i(t) = 3 + 5t$, where *i* is in amperes & t is in seconds. Take R = 4Ω, L = 6H & find an expression for the battery emf as function of time.



33. A rectangular frame ABCD made of a uniform metal wire has a straight connection between E & F made of the same wire as shown in the figure. AEFD is a square of side 1 m & EB = FC = 0.5 m. The entire circuit is placed in a steadily increasing uniform magnetic field directed into the place of the paper & normal to it . The rate of change of the magnetic field is 1 T/s, the resistance per unit length of the wire is 1 Ω/m. Find the current in segments AE, BE & EF.



34. Two parallel vertical metallic rails AB & CD are separated by 1 m. They are connected at the two ends by resistance R_1 & R_2 as shown in the figure. A horizontally metallic bar L of mass 0.2 kg slides without friction, vertically down the rails under the action of gravity. There is a uniform horizontal magnetic field of 0.6T perpendicular to the plane of the rails, it is observed that when the terminal velocity is attained, the power dissipated in R_1 & R_2 are 0.76 W & 1.2 W respectively. Find the terminal velocity of bar L & value R_1 & R_2. (Take g = 9.8 m/s^2)

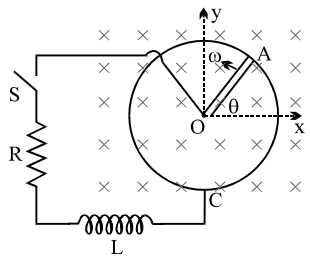


35. A metal rod OA of mass m & length r is kept rotating with a constant angular speed ω in a vertical plane about a horizontal axis at the end O. The free end A is arranged to slide without friction along a fixed conducting circular ring in the same plane as that of rotation. A uniform & constant magnetic induction $\vec{B}$ is applied perpendicular & into the plane of rotation as shown in figure. An inductor L and an external resistance R are connected through a switch S between the point O & a point C on the ring to form an electrical circuit. Neglect the resistance of the ring and the rod. Initially, the switch is open.

(a) What is the induced emf across the terminals of the switch ?

(b) (i) Obtain an expression for the current as a function of time after switch S is closed.

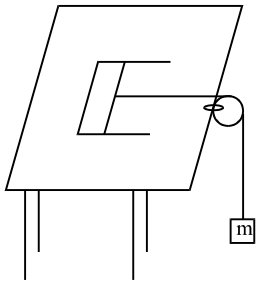
(ii) Obtain the time dependence of the torque required to maintain the constant angular speed, given that the rod OA was along the positive X-axis at t = 0.



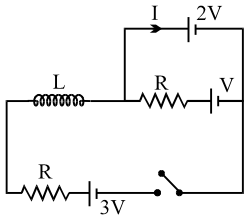
36. A pair of parallel horizontal conducting rails of negligible resistance shorted at one end is fixed on a table. The distance between the rails is L. A conducting massless rod of resistance R can slide on the rails frictionlessly. The rod is tied to a massless string which passes over a pulley fixed to the edge of the table. A mass m, tied to the other end of the string hangs vertically. A constant magnetic field B exists perpendicular to the table. If the system is released from rest, calculate :

(i) the terminal velocity achieved by the rod.

(ii) the acceleration of the mass at the instant when the velocity of the rod is half the terminal velocity.



37. In the LR circuit shown, what is the variation of the current I as a function of time? The switch is closed at time t = 0 sec.



38. A current of 4 A flows in a coil when connected to a 12 V dc source. If the same coil is connected to a 12V, 50 rad/s ac source a current of 2.4 A flows in the circuit. Determine the inductance of the coil. Also find the power developed in the circuit if a 2500 μF capacitor is connected in series with the coil.

39. An LCR series circuit with 100Ω resistance is connected to an ac source of 200 V and angular frequency 300 rad/s. When only the capacitance is removed, the current lags behind the voltage by 60°. When only the inductance is removed, the current leads the voltage by 60°. Calculate the current and the power dissipated in the LCR circuit.

40. A box P and a coil Q are connected in series with an ac source of variable frequency. The emf of source at 10 V. Box P contains a capacitance of 1μF in series with a resistance of 32Ω coil Q has a self-inductance 4.9 mH and a resistance of 68Ω series. The frequency is adjusted so that the maximum current flows in P and Q. Find the impedance of P and Q at this frequency. Also find the voltage across P and Q respectively.

41. A series LCR circuit containing a resistance of 120Ω has angular resonance frequency 4×10^5 rad s^{-1}. At resonance the voltages across resistance and inductance are 60 V and 40 V respectively. Find the values of L and C. At what frequency the current in the circuit lags the voltage by 45°?

42. An inductor of inductance 2.0 mH, is connected across a charged capacitor of capacitance 5.0 μF and the resulting LC circuit is set oscillating at its natural frequency. Let Q denote the instantaneous charge on the capacitor and I the current in the circuit. It is found that the maximum value of Q is 200 μF.

(a) when Q = 100 μC, what is the value of |dI / dt| ?

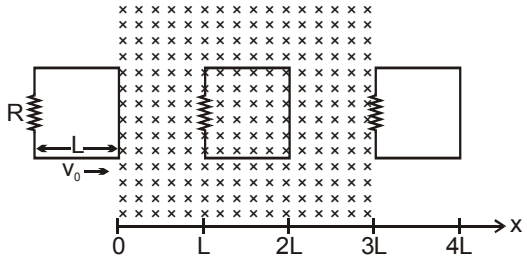
(b) when Q = 200 μC, what is the value of I?

(c) Find the maximum alue of I.

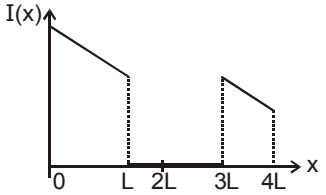
(d) when I is equal to one half its maximum value, what is the value of |Q|

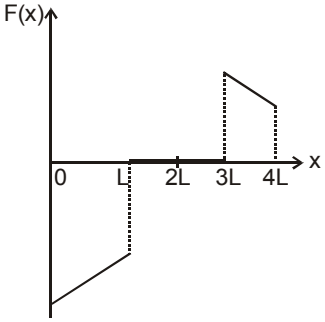
EXERCISE-III

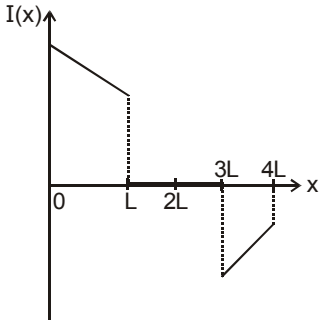
1. A rigid wire loop of square shape having side of length L and resistance R is moving along the x-axis with a constant velocity v_0 in the plane of the paper. At t = 0, the right edge of the loop enters a region of length 3L where there is a uniform magnetic field B_0 into the plane of the paper, as shown in the figure. For sufficiently large v_0, the loop eventually crosses the region. Let x be the location of the right edge of the loop. Let v(x), I(x) and F(x) represent the velocity of the loop, current in the loop, and force on the loop, respectively, as a function of x. Counter-clockwise current is taken as positive. **[JEE Advanced-2016]**

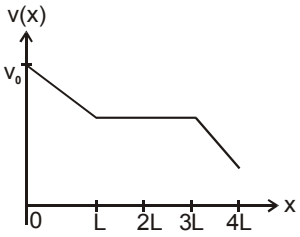


Which of the following schematic plot(s) is(are) correct ? (Ignore gravity)

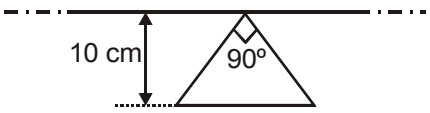
(A)


(B)


(C)


(D)


2. A conducting loop in the shape of a right angled isosceles triangle of height 10 cm is kept such that the 90° vertex is very close to an infinitely long conducting wire (see the figure). The wire is electrically insulated from the loop. The hypotenuse of the triangle is parallel to the wire. The current in the triangular loop is in counterclockwise direction and increased at a constant rate of 10 As^{-1}. Which of the following statement(s) is(are) true? **[JEE Advanced-2016]**



(A) The induced current in the wire is in opposite direction to the current along the hypotenuse

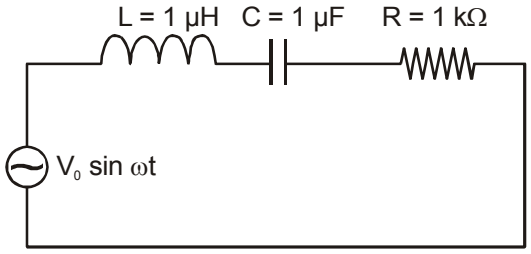
(B) The magnitude of induced emf in the wire is $\left(\frac{\mu_0}{\pi}\right)$ volt

(C) There is a repulsive force between the wire and the loop

(D) If the loop is rotated at a constant angular speed about the wire, an additional emf of $\left(\frac{\mu_0}{\pi}\right)$ volt is induced in the wire

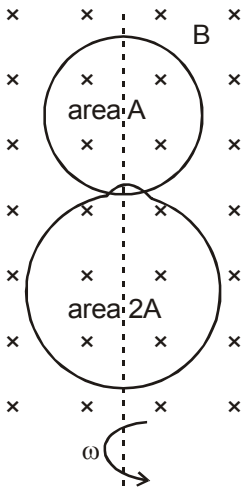
3. Two inductors L_1 (inductance 1 mH, internal resistance 3Ω) and L_2 (inductance 2 mH, internal resistance 4Ω), and a resistor R (resistance 12Ω) are all connected in parallel across a 5 V battery. The circuit is switched on at time t = 0. The ratio of the maximum to the minimum current (I_{max} / I_{min}) drawn from the battery is **[JEE Advanced-2016]**

4. In the circuit show, L = 1 μH, C = 1 μF and R = 1 kΩ. They are connected in series with an a.c. source $V = V_0 \sin \omega t$ as shown. Which of the following options is/are correct ? **[JEE Advanced-2017]**



(A) At $\omega >> 10^6$ rad. s^{-1}, the circuit behaves like a capacitor
(B) The frequency at which the current will be in phase with the voltage is independent of R
(C) The current will be in phase with the voltage if $\omega = 10^4$ rad. s^{-1}
(D) At $\omega \sim 0$ the current flowing through the circuit becomes nearly zero

5. A circular insulated copper wire loop is twisted to form two loops of area A and 2A as shown in the figure. At the point of crossing the wires remain electrically insulated from each other. The entire loop lies in the plane (of the paper). A uniform magnetic field $\vec{B}$ points into the plane of the paper. At t = 0, the loop starts rotating about the common diameter as axis with a constant angular velocity ω in the magnetic field. Which of the following options is/are correct ? **[JEE Advanced-2017]**



(A) The net emf induced due to both the loops is proportional to cos ωt
(B) The amplitude of the maximum net emf induced due to both the loops is equal to the amplitude of maximum emf induced in the smaller loop alone
(C) The rate of change of the flux is maximum when the plane of the loops is perpendicular to plane of the paper
(D) The emf induced in the loop is proportional to the sum of the areas of the two loops

6. The instantaneous voltages at three terminals marked X, Y and Z are given by $V_X = V_0 \sin \omega t$, $V_Y = V_0 \sin\left(\omega t + \frac{2\pi}{3}\right)$ and $V_Z = V_0 \sin\left(\omega t + \frac{4\pi}{3}\right)$

an ideal voltmeter is configured to read rms value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading(s) of the voltmeter will be

[JEE Advanced-2017]

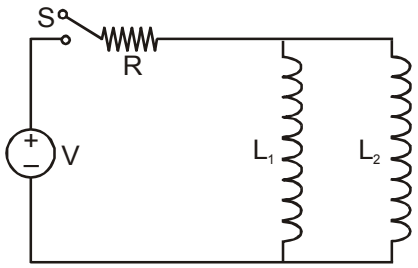
(A) $V_{XY}^{rms} = V_0$

(B) $V_{YZ}^{rms} = V_0\sqrt{\frac{1}{2}}$

(C) independent of the choice of the two terminals

(D) $V_{XY}^{rms} = V_0\sqrt{\frac{3}{2}}$

7. A source of constant voltage V is connected to a resistance R and two ideal inductors L_1 and L_2 through a switch S as shown. There is no mutual inductance between the two inductors. The switch S is initially open. At t = 0, the switch is closed and current begins to flow. Which of the following options is/are correct?



(A) The ratio of the currents through L_1 and L_2 is fixed at all times (t > 0) **[JEE Advanced-2017]**

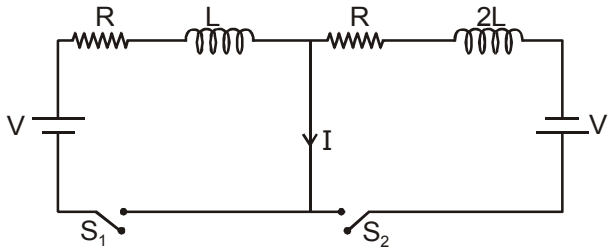
(B) After a long time, the current through L_1 will be $\frac{V}{R}\frac{L_2}{L_1+L_2}$

(C) After a long time, the current through L_2 will be $\frac{V}{R}\frac{L_1}{L_1+L_2}$

(D) At t = 0, the current through the resistance R is $\frac{V}{R}$

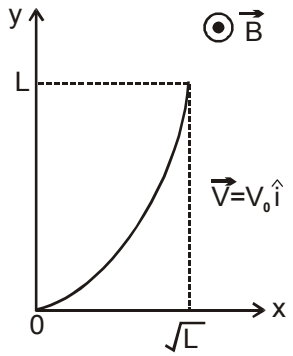
8. In the figure below, the switches S_1 and S_2 are closed simultaneously at t = 0 and a current starts to flow in the circuit. Both the batteries have the same magnitude of the electromotive force (emf) and the polarities are as indicated in the figure. Ignore mutual inductance between the inductors. The current I in the middle wire reaches its maximum magnitude I_{max} at time $t = \tau$. Which of the following statements is(are) ture?

[JEE-Advanced-2018]



(A) $I_{max} = \frac{V}{2R}$ (B) $I_{max} = \frac{V}{4R}$ (C) $\tau = \frac{L}{R}\ln 2$ (D) $\tau = \frac{2L}{R}\ln 2$

9. A conducting wire of parabolic shape, initially $y = x^2$, is moving with velocity $\vec{V} = V_0\hat{i}$ in a non-uniform magnetic field $\vec{B} = B_0\left(1+\left(\frac{y}{L}\right)^{\beta}\right)\hat{k}$, as shown in figure. If V_0, B_0, L and β are positive constants and $\Delta\phi$ is the potential difference developed between the ends of the wire, then the correct statement(s) is/are :



(A) $|\Delta\phi| = \frac{1}{2}B_0V_0L$ for $\beta = 0$ **[JEE-Advanced-2019]**

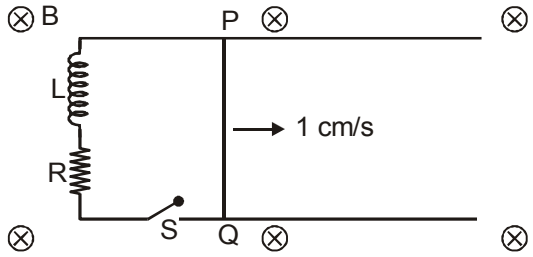
(B) $|\Delta\phi|$ is proportional to the length of the wire projected on the y-axis.

(C) $|\Delta\phi|$ remains the same if the parabolic wire is replaced by a straight wire. y = x initially, of length $\sqrt{2}L$

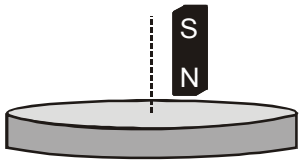
(D) $|\Delta\phi| = \frac{4}{3}B_0V_0L$ for $\beta = 2$

10. A 10 cm long perfectly conducting wire PQ is moving with a velcoity 1 cm/s on a pair of horizontal rails of zero resistance. One side of the rails is connected to an inductor L = 1 mH and a resistance R = 1Ω as shown in figure. The horizontal rails, L and R lie in the same plane with a uniform magnetic field B = 1 T perpendicular to the plane. If the key S is closed at certain instant, the current in the circuit after 1 millisecond is $x \times 10^{-3}$ A, where the value of x is _____.

[Assume the velocity of wire PQ remains constant (1 cm/s) after key S is closed. Given : $e^{-1} = 0.37$, where e is base of the natural logarithm] **[JEE-Advanced-2019]**



11. A light disc made of aluminium (a nonmagnetic material) is kept horizontally and is free to rotate about its axis as shown in the figure. A strong magnet is held vertically at a point above the disc away from its axis. On revolving the magnet about the axis of the disc, the disc will (figure is schematic and not drawn to scale)



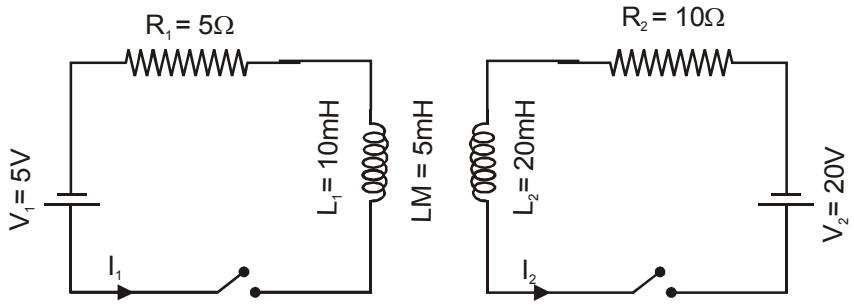
(A) rotate in the direction opposite to the direction of magnet's motion **[JEE-Advanced-2020]**

(B) rotate in the same direction as the direction of magnet's motion

(C) not rotate and its temperature will remain unchanged

(D) not rotate but its temperature will slowly rise

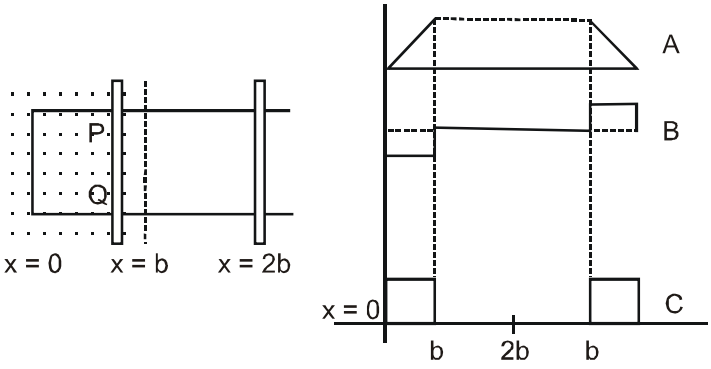
12. The inductors of two LR circuits are placed next to each other, as shown in the figure. The value of the self-inductance of the inductors, resistances, mutual-inductance and applied voltage are specified in the given circuit. After both the switches are closed simultaneously, the total work done by the batteries against the induced EMF in the inductors by the time the currents reach their steady state values is _____mJ. **[JEE-Advanced-2020]**



13. A common transistor radio set requires 12 V (D.C.) for its operation. The D.C. source constructed by using a transformer and a rectifier circuit, which are operated at 220 (A.C) on standard domestic A.C. supply. The number of turns of secondary coil are 24, then the number of turns of primary are. **[JEE-Main-Online-2021]**

14. The arm PQ of a rectangular conductor is moving from x = 0 to x = 2b outwards and then inwards from x = 2b to x = 0 as shown in the figure. A uniform magnetic field perpendicular to the plane is acting from x = 0 to x = b. Identify the graph showing the variation of different quantities with distance.

[JEE-Main-Online-2021]



(A) A-Power dissipated, B-Flux, C-EMF (B) A-EMF B-Power dissipated, C-Flux

(C) A-Flux,B-Power dissipated, C-EMF (D) A-Flux, B-EMF, C-Power dissipated

15. A 0.07 H inductor and a 12Ω resistance are connected in series to a 220 V, 50 Hz ac source.The approximate current in the circuit and the phase angle between current and source voltage are respectively.

[Take π as $\frac{22}{7}$] **[JEE-Main-Online-2021]**

(A) 8.8 A and $\tan^{-1}\left(\frac{6}{11}\right)$ (B) 8.8 A and $\tan^{-1}\left(\frac{11}{6}\right)$

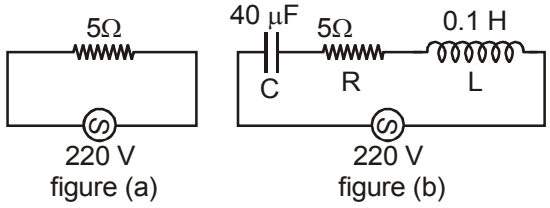
(C) 0.88 A and $\tan^{-1}\left(\frac{11}{6}\right)$ (D) 88 A and $\tan^{-1}\left(\frac{11}{6}\right)$

16. A 100 Ω resistance, a 0.1 μF capacitor and an inductor are connected in series across a 250 V supply at variable frequency. Calculate the value of inductance of inductor at which resonance will occur. Given that the resonant frequency is 60 Hz. **[JEE-Main-Online-2021]**

(A) 70.3 mH (B) 7.03×10^{-5} H (C) 70.3 H (D) 0.70 H

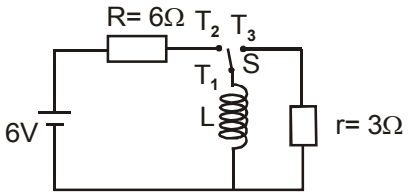
17. A circular conducting coil of radius 1 m is being heated by the change of magnetic field $\vec{B}$ passing perpendicular to the plane in which the coil is laid. The resistance of the coil is 2 μΩ. The magnetic field is slowly switched off such that its magnitude changes in time as B = $\frac{4}{\pi}\times10^{-3}\,T\left(1-\frac{t}{100}\right)$. The energy dissipated by the coil before the magnetic field is switched off completely is E = _____ mJ. **[JEE-Main-Online-2021]**

18. Two circuits are shown in the figure (a) & (b). At a frequency of _____rad/s the average power dissipated in one cycle will be same in both the circuits. **[JEE-Main-Online-2021]**



19. An inductor of 10 mH is connected to a 20 V battery through a resistor of 10 kΩ and a switch. After a long time, when maximum current is set up in the circuit, the current is switched off. The current in the circuit after 1 μs is $\frac{x}{100}$ mA. Then x is equal to_____. (Take $e^{-1} = 0.37$) **[JEE-Main-Online-2021]**

20. Consider an electrical circuit containing a two way switch 's'. Initially S is open and then T_1 is connected to T_2 . As the current in R = 6Ω attains a mixture value of steady steady state level, T_1 is disconnected from T_2 and immediately connected to T_3. Potential drop across r = 3Ω resistor immediately after T_1 is connected to T_3 is ..... V. (Round off to the Nearest integer) **[JEE-Main-Online-2021]**



21. **Match List-I with List-II :** **[JEE-Main-Online-2021]**

	List-I		**List-II**
(a)	$\omega L > \frac{1}{\omega C}$	(i)	Current is in phase with emf
(b)	$\omega L = \frac{1}{\omega C}$	(ii)	Current lags behind the applied emf
(c)	$\omega L < \frac{1}{\omega C}$	(iii)	Maximum current occurs
(d)	Resonant frequency	(iv)	Current leads the emf

Choose the **correct** answer from the options given below :

(A) (a)-(iv); (b)-(iii); (c)-(ii); (d)-(i) (B) (a)-(ii); (b)-(i); (c)-(iv); (d)-(iii)

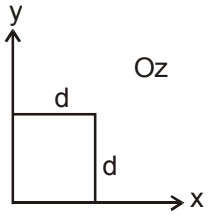
(C) (a)-(iii); (b)-(i); (c)-(iv); (d)-(ii) (D) (a)-(ii); (b)-(i); (c)-(iii); (d)-(iv)

22. For a series LCR circuit with R = 100 Ω, L = 0.5 mH and C = 0.1 pF connected across 220 V–50 Hz AC supply, the phase angle between current and supplied voltage and the nature of the circuit is : **[JEE-Main-Online-2021]**

(A) 0°, resonance circuit (B) 0°, resistive circuit

(C) ≈ 90°, predominantly inductive circuit (D) ≈ 90°, predominantly capactive circuit

23. The magnetic field in a region is given by $\vec{B} = B_0\left(\frac{x}{a}\right)\hat{k}$. A square loop of side d is placed with its edges along the x and y axes. The loop is moved with a constant velocity $\vec{v} = v_0\hat{i}$ The emf inductance in the loop is : **[JEE-Main-Online-2021]**



(A) $\frac{B_0 v_0^2 d}{2a}$ (B) $\frac{B_0 v_0 d}{2a}$ (C) $\frac{B_0 v_0 d^2}{a}$ (D) $\frac{B_0 v_0 d^2}{2a}$

24. Match List-I with List-II **[JEE-Main-Online-2021]**

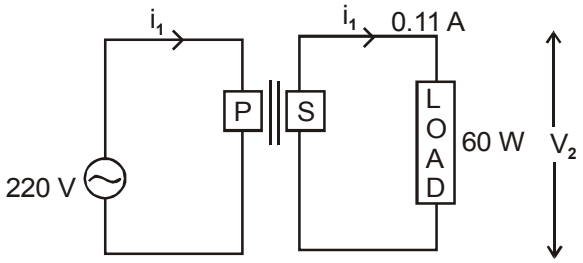
	List-I		List-II
(a)	Phase difference between current and voltage in a purely resistive AC circuit	(i)	$\frac{\pi}{2}$; current leads voltage
(b)	Phase difference between current and voltage in a pure inductive AC circuit	(ii)	zero
(c)	Phase difference between current and voltage in a pure capacity AC circuit	(iii)	$\frac{\pi}{2}$; current lags voltage
(d)	Phase difference between current and voltage in an LCR series circuit	(iv)	$\tan^{-1}\left(\frac{X_C - X_L}{R}\right)$

Choose the most appropriate answer from the options given below :

(A) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii) (B) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(C) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) (D) (a)-(ii), (b) (iii), (c)-(i), (d)-(iv)

25. For the given circuit, comment on the type of transformer used. **[JEE-Main-Online-2021]**



(A) Auxilliary transformer (B) Auto transformer

(C) Step -up transformer (D) Step down transformer

26. In a seires LCR resonance circuit, if we charge the resistance only, from a lower to higher value :

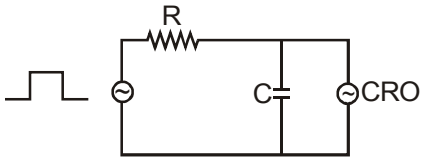
(A) The bandwidth of resonance circuit will increase **[JEE-Main-Online-2021]**

(B) The resonance frequency will increase

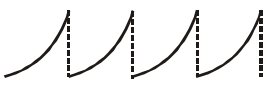
(C) The quality factor will incrase

(D) The quality factor and the resonance frequency will remian constant

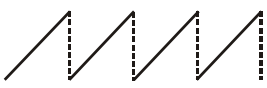
27. An RC circuit as shown in the figure is driven by a AC source generating a square wave. The output wave pattern monitored by CRO would look close to : **[JEE-Main-Online-2021]**



(A) 

(B) 

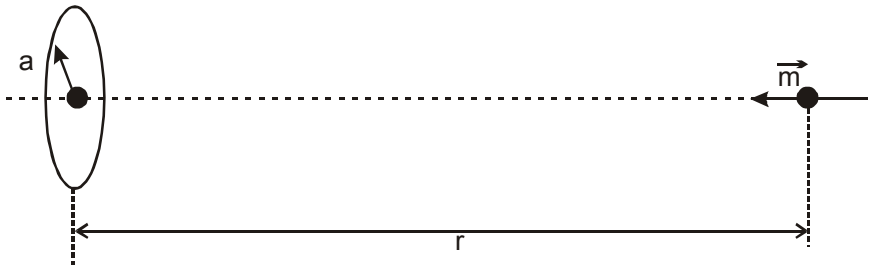
(C) 

(D) 

Paragraph for Question Nos. 28 to 29

A special metal S conducts electricity without any resistance. A closed wire loop, made of S, does not allow any change in flux through itself by inducing a suitable current to generate a compensating flux. The induced current in the loop cannot decay due to its zero resistance. The current gives rise to a magnetic moment which in turn repels the source of magnetic field or flux. Consider such a loop, of radius a, with its center at the origin. A magnetic dipole of moment m is brought along the axis of this loop from infinity to a point at distance r (>> a) from the center of the loop with its north pole always facing the loop, as shown in the figure below.

The magnitude of magnetic field of a dipole m, at a point on its axis at distance r, is $\frac{\mu_0}{2\pi}\frac{m}{r^3}$, where μ_0 is the parmeability of free space. The magnitude of the force between two magnetic dipoles with moments, m_1 and m_2, separated by a distance r on the common axis, with their north poles facing each other, is $\frac{km_1m_2}{r^4}$, where k is a constant of appropriate dimensions. The direction of this force is along the line joining the two dipoles. **[JEE Advance-2021]**



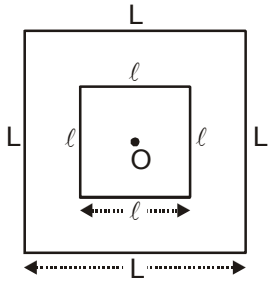
28. When the dipole m is placed at a distance r from the center of the loop (as shown in the figure), the current induced in the loop will be proportional to

(A) $\frac{m}{r^3}$ (B) $\frac{m^2}{r^2}$ (C) $\frac{m}{r^2}$ (D) $\frac{m^2}{r}$

29. The work done in bringing the dipole from infinity to a distance r from the center of the loop by the given process is proportional to

(A) $\frac{m}{r^5}$ (B) $\frac{m^2}{r^5}y$ (C) $\frac{m^2}{r^6}$ (D) $\frac{m^2}{r^7}$

30. A small square loop of wire of side ℓ is placed inside a large square loop of wire L (L >> ℓ). Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is **[JEE-Main-Online-2022]**



(A) $\frac{2\sqrt{2}\mu_0L^2}{\pi\ell}$ (B) $\frac{\mu_0\ell^2}{2\sqrt{2}\pi L}$ (C) $\frac{2\sqrt{2}\mu_0\ell^2}{\pi L}$ (D) $\frac{\mu_0L^2}{2\sqrt{2}\pi\ell}$

31. A coil of inductance 1 H and resistance 100 Ω is connected to a battery of 6 V. Determine approximately :
(a) The time elapsed before the current acquires half of its steady – state value
(b) The energy stored in the magnetic field associated with the coil at an instant 15 ms after the circuit is switched on. (Given ln2 = 0.693, $e^{-3/2} = 0.25$) **[JEE-Main-Online-2022]**
(A) t = 10 ms; U = 2 mJ (B) t = 10 ms; U = 1 mJ
(C) t = 7 ms; U = 1 mJ (D) t = 7 ms; U = 2 mJ

32. The magnetic moment of an electron(e) revolving in an orbit around nucleus with an orbital angular momentum is given by **[JEE-Main-Online-2022]**

(A) $\vec{\mu}_L = \frac{e\vec{L}}{2m}$ (B) $\vec{\mu}_L = -\frac{e\vec{L}}{2m}$ (C) $\vec{\mu}_1 = -\frac{e\vec{L}}{m}$ (D) $\vec{\mu}_1 = \frac{2e\vec{L}}{m}$

33. A capacitor of capacitance 500 μF is charged completely using a dc supply of 100 V. It is now connected to an inductor of inductance 50 mH to form an LC circuit. The maximum current in LC circuit will be_____A. **[JEE-Main-Online-2022]**

34. An alternating emf E = 440 sin 100πt is applited to a circuit containing an inductance of $\frac{\sqrt{2}}{\pi}$H. If an a.c. ammeter is connected in the circuit, its reading will be : **[JEE-Main-Online-2022]**
(A) 4.4 A (B) 1.55 A (C) 2.2 A (D) 3.11 A

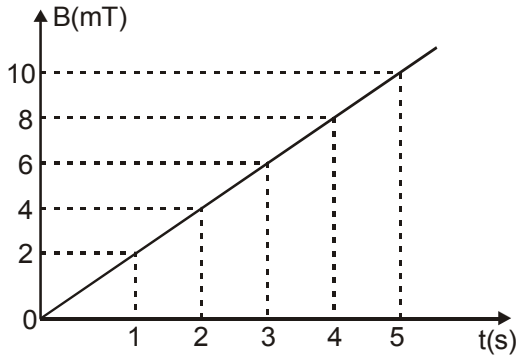
35. Consider an LC circuit, with inductance L = 0.1 H and capacitance C = 10^{-3}F, kept on a plane. The area of the circuit is 1 m^2. It is placed in a constant magnetic field of strength B_0 which is perpendicular to the plane of the circuit. At time t = 0, the magnetic field strength starts increasing linearly as $B = B_0 + \beta t$ with $\beta = 0.04\ Ts^{-1}$. The maximum magnitude of the current in the circuit is _____ mA. **[JEE-Advanced-2022]**

36. A 12 V battery connected to a coil of resistance 6 Ω through a switch, drives a constant current in the circuit. The switch is opened in 1 ms. The emf induced across the coil is 20 V. The inductance of the coil is **[JEE-Main-Online-2023]**
(A) 8 mH (B) 10 mH (C) 5 mH (D) 12 mH

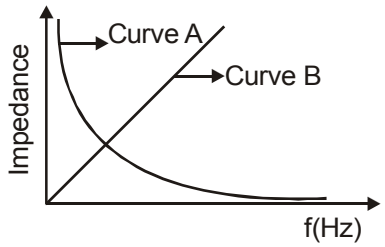
37. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R
Assertion (A) : A bar magnet dropped through a metallic cylindrical pipe takes more time to come down compared to a non-magnetic bar with same geometry and mass.
Reason (R) : For the magnetic bar, Eddy currents are produced in the metallic pipe which oppose the motion of the magnetic bar.
In the light of the above statement, choose the correct answer from the options given below :
(A) Both A and R are true and R is the correct explanation of A
(B) A is false but R is true
(C) Both A and R are true but R is NOT the correct explanation of A **[JEE-Main-Online-2023]**
(D) A is true but R is false

38. The magnetic field B crossing normally a square metallic plate of area 4 m^2 is changing with time as shown in figure. The magnitude of induced emf in the plate during t = 2s to t = 4s, is _____mV.

[JEE-Main-Online-2023]



39. As per the given graph, choose the correct representation for curve A and curve B.



{Where X_C = reactance of pure capacitive circuit connected with A.C. source
X_L = reactance of pure inductive circuit connected with A.C. source.
R = impedance of pure resistive circuit connected with A.C. source.
Z = Impedance of the LCR series circuit} **[JEE-Main-Online-2023]**

(A) $A = X_L, B = Z$ (B) $A = X_C, B = R$ (C) $A = X_L, B = R$ (D) $A = X_C, B = X_L$

40. An ideal transformer with purely resistive load operates at 12 kV on the primary side. It supplies electrical energy to a number of nearby houses at 120 V. The average rate of energy consumption in the houses served by the transformer is 60 kW. The value of resistive load (Rs) required in the secondary circuit will be_____mΩ. **[JEE-Main-Online-2023]**

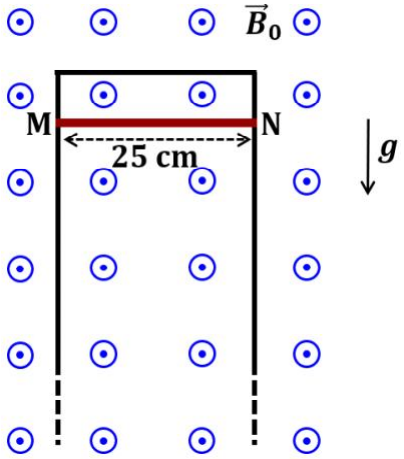
41. A series LCR circuit is connected to a 45 sin(ωt) Volt source. The resonant angular frequency of the circuit is 10^5 rad s^{-1} and current amplitude at resonance is I_0. When the angular frequency of the source is $\omega = 8 \times 10^4$ rad s^{-1}, the current amplitude in the circuit is 0.05 I_0. If L = 50 mH, match each entry in List-I with an appropriate value from List-II and choose the correct option. **[JEE-Advanced-2023]**

	List-I		**List-II**
(P)	I_0 in mA	(1)	44.4
(Q)	The quality factor of the circuit	(2)	18
(R)	The bandwidth of the circuit in rad s^{-1}	(C)	400
(S)	The peak power dissipated at resonance in Watt	(D)	2250
		(5)	500

(A) P $\rightarrow$ 2, Q $\rightarrow$ 3, R $\rightarrow$ 5, S $\rightarrow$ 1 (B) P $\rightarrow$ 3, Q $\rightarrow$ 1, R $\rightarrow$ 4, S $\rightarrow$ 2
(C) P $\rightarrow$ 4, Q $\rightarrow$ 5, R $\rightarrow$ 3, S $\rightarrow$ 1 (D) P $\rightarrow$ 4, Q $\rightarrow$ 2, R $\rightarrow$ 1, S $\rightarrow$ 5

42. A thin conducting rod MN of mass 20 gm, length 25 cm and resistance 10 Ω is held on frictionless, long, perfectly conducting vertical rails as shown in the figure. There is a uniform magnetic field B_0 = 4 T directed perpendicular to the plane of the rod-rail arrangement. The rod is released from rest at time t = 0 and it moves down along the rails. Assume air drag is negligible. Match each quantity in List-I with an appropriate value from List-II, and choose the correct option.

[Given : The acceleration due to gravity g = 10 m s^{-2} and e^{-1} = 0.4] **[JEE-Advanced-2023]**

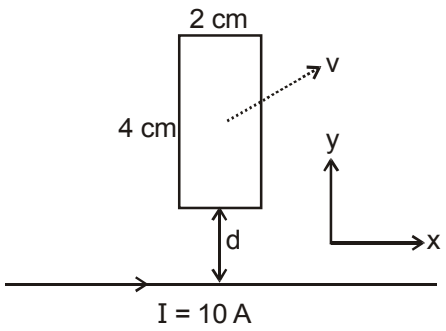


	List-I		List-II
(P)	At t = 0.2 s, the magnitude of the induced emf in Volt	(1)	0.07
(Q)	At t = 0.2 s, the magnitude of the magnetic force in Newton	(2)	0.14
(R)	At t = 0.2 s, the power dissipated as heat in Watt	(C)	1.20
(S)	The magnitude of terminal velocity of the rod in m s^{-1}	(D)	0.12
		(5)	2.00

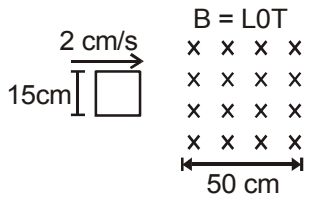
(A) P → 5, Q → 2, R → 3, S → 1 (B) P → 3, Q → 1, R → 4, S → 5

(C) P → 4, Q → 3, R → 1, S → 2 (D) P → 3, Q → 4, R → 2, S → 5

43. A rectangular conducting loop of length 4 cm and width 2 cm is in the xy-plane, as shown in the figure. It is being moved away from a thin and long conducting wire along the direction $\frac{\sqrt{3}}{2}\hat{x}+\frac{1}{2}\hat{y}$ with a constant speed v. The wire is carrying a steady current I = 10 A in the positive x-direction. A current of 10 μA flows through the loop when it is at a distance d = 4 cm from the wire. If the resistance of the loop is 0.1 Ω, then the value of v is___m s^{-1}. [Given: The permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ N A^{-2}] **[JEE-Advanced-2023]**



44. A square loop of side 15 cm being moved towards right at a constant speed of 2 cm/s as shown in figure. The front edge enters the 50 cm wide magnetic field at t = 0. The value of induced emf in the loop at t = 10 s will be : **[JEE-Main-Online-2024]**

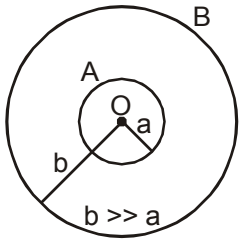


(A) 3 mV (B) 4.5 mV (C) 0.3 mV (D) zero

45. In a coil, the current changes from –2 A to +2A in 0.2 s and induces an emf of 0.1 V. The self inductance of the coil is : **[JEE-Main-Online-2024]**

(A) 1 mH (B) 4 mH (C) 2.5 mH (D) 5 mH

46. Two conducting circular loops A and B are placed in the same plane with their centres coinciding as shown in figure. The mutual inductance between them is **[JEE-Main-Online-2024]**

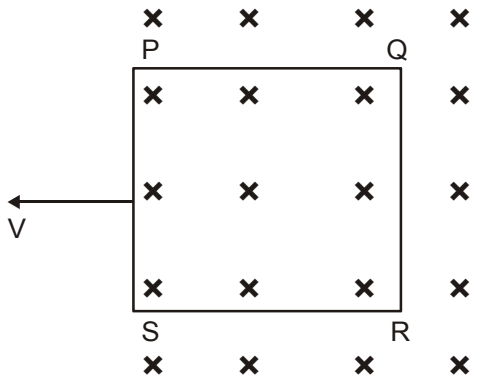


(A) $\frac{\mu_0}{2\pi} \cdot \frac{b^2}{a}$ (B) $\frac{\mu_0}{2\pi} \cdot \frac{a^2}{b}$ (C) $\frac{\mu_0 \pi b^2}{2a}$ (D) $\frac{\mu_0 \pi a^2}{2b}$

47. A rod of length 60 cm rotates with a uniform angular velocity 20 rad s^{-1} about its perpendicular bisector, in a uniform magnetic field 0.5 T. The direction of magnetic field is parallel to the axis of rotation. The potential difference between the two ends of the rod is_____V. **[JEE-Main-Online-2024]**

48. The current in an inductor is given by I = (3t + 8) where t is in second. The magnitude of induced emf produced in the inductor is 12 mV. The self-inductance of the inductor_____mH. **[JEE-Main-Online-2024]**

49. A square loop PQRS having 10 turns, area 3.6×10^{-3} m^2 and resistance 100 Ω is slowly and uniformly being pulled out of a uniform magnetic field of magnitude B = 0.5 T as shown. Work done in pulling the loop out of the field in 1.0 s is _____ × 10^{-6} J. **[JEE-Main-Online-2024]**



50. A coil of negligble resistance is connected in series with 90 Ω resistor across 120 V, 60 Hz supply. A voltmeter reads 36 V across resistance. Inductance of the coll is : **[JEE-Main-Online-2024]**

(A) 0.76 H (B) 0.91 H (C) 2.86 H (D) 0.286 H

51. A LCR circuit is at resonance for a capacitor C, inductance L and resistance R. Now the value of resistance is halved keeping all other parameters same. The current amplitude at resonance will be now : **[JEE-Main-Online-2024]**

(A) Zero (B) Halve (C) Double (D) Same

52. Given below are two statements : **[JEE-Main-Online-2024]**

Statement-I : In an LCR series circuit, current is maximum at resonance.

Statement-II : Current in a purely resistive circuit can never be less than that in a series LCR circuit when connected to same voltage source.

In the light of the above statements, choose the correct from the options given below :

(A) Both Statement I and Statement II are true (B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false (D) Statement I is true but Statement II is false

53. In an ac circuit, the instantaneous current is zero, when the instantaneous voltage is maximum. In this case, the source may be connected to **[JEE-Main-Online-2024]**

(1) Pure inductor (2) Pure capacitor

(C) Pure resistor (D) Combination of an inductor and capacitor

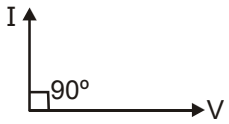
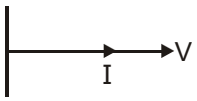
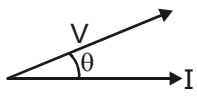
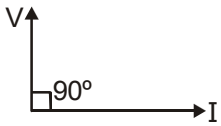
Choose the **correct** answer from the options given below

(A) 2, 3 and 4 only (B) 1 and 2 only (C) 1, 2 and 4 only (D) 1, 2 and 3 only

54. A series LCR circuit is subjected to an ac signal of 200 V, 50 Hz. If the voltage across the inductor (L = 10 mH) is 31.4 V, then the current in this circuit is_____. **[JEE-Main-Online-2024]**

(A) 10 A (B) 10 mA (C) 68 A (D) 63 A

55. Match List-I with List-II

	List-I		**List-II**
(a)	Purely capacitive circuit	(I)	
(b)	Purely inductive circuit	(II)	
(c)	LCR series at resonance	(III)	
(d)	LCR series circuit	(IV)	

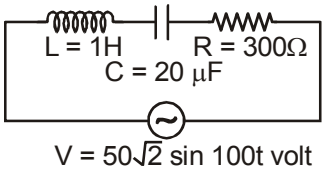
Choose the correct answer from the options given below : **[JEE-Main-Online-2024]**

(A) a-I, b-IV, c-II, d-III (B) a-IV, b-I, c-III, d-II

(C) a-I, b-IV, c-III, d-II (D) a-IV, b-I, c-II, d-III

56. A alternating current at any instant is given by $i = \left[6 + \sqrt{56}\sin(100\pi t + \pi/3)\right]$A. The rms value of the current is_____A. **[JEE-Main-Online-2024]**

57. An ac source is connected in given series LCR circuit. The rms potential difference across the capacitor of 20 μF is_____V. **[JEE-Main-Online-2024]**



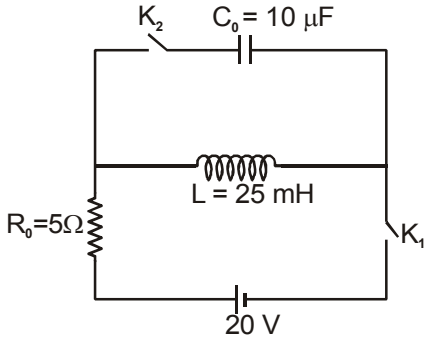
58. When a dc voltage of 100V is applied to an inductor, a dc current of 5A flows through it. When an ac voltage of 200V peak value is connected to inductor, its inductive reactance is found to be $20\sqrt{3}\,\Omega$. The power dissipated in the circuit is_____W. **[JEE-Main-Online-2024]**

59. For a given series LCR circuit it is found that maximum current is drawn when value of variable capacitance is 2.5 nF. If resistance 200Ω and 100 mH inductor is being used in the given circuit. The frequency of ac source is _____ × 10^3 Hz. (given π^2 = 10) **[JEE-Main-Online-2024]**

60. An alternating e.m.f. E = 110 $\sqrt{2}$ sin100t volt is applied to a capacitor of 2μF, the rms value of current in the circuit is _____ mA. **[JEE-Main-Online-2024]**

61. When a coil is connected across a 20 V dc supply, it draws a current of 5 A. When it is connected across 20 V, 50 Hz ac suplly, it draws a current of 4 A. The self inductance of the coil is _____mH. (Take π = 3) **[JEE-Main-Online-2024]**

62. The circuit shown in the figure contains an inductor L, a capacitor C_0, a resistor R_0 and an ideal battery. The circuit also contains two keys K_1 and K_2. Initially, both the keys are open and there is no charge on the capacitor. At an instant, key K_1 is closed and immediately after this the current in R_0 is found to be I_1. After a long time, the current attains a steady state value I_2. Thereafter, K_2 is closed and simultaneously K_1 is opened and the voltage across C_0 oscillates with amplitude V_0 and angular frequency ω_0.



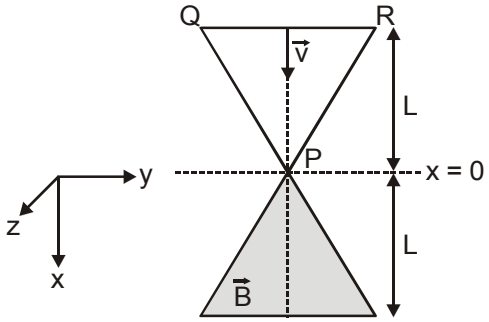
Match the quantities mentioned in List-I with their values in List-II and choose the correct option. **[JEE-Advanced-2024]**

	List-I		**List-II**
(P)	The value of I_1 in Ampere is	(1)	0
(Q)	The value of I_2 in Ampere is	(2)	2
(R)	The value of ω_0 in kilo-radians/s is	(C)	4
(S)	The value of V_0 in Volt is	(D)	20
		(5)	200

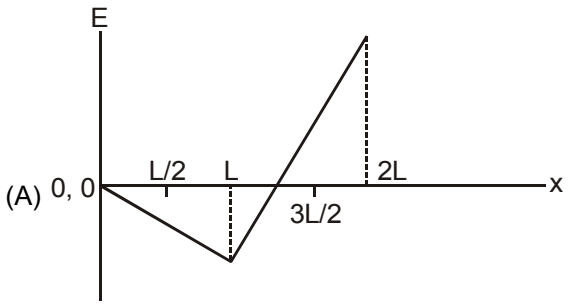
(A) P → 1 ; Q → 3 ; R → 2 ; S → 5 (B) P → 1 ; Q → 2 ; R → 3 ; S → 5

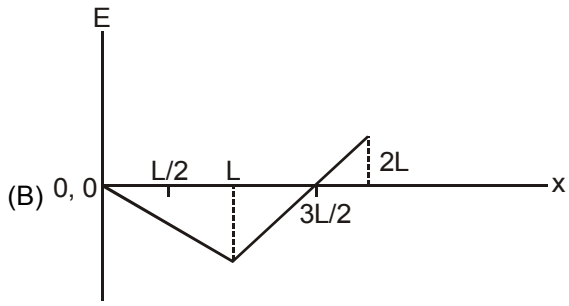
(C) P → 1 ; Q → 3 ; R → 2 ; S → 4 (D) P → 2 ; Q → 5 ; R → 3 ; S → 4

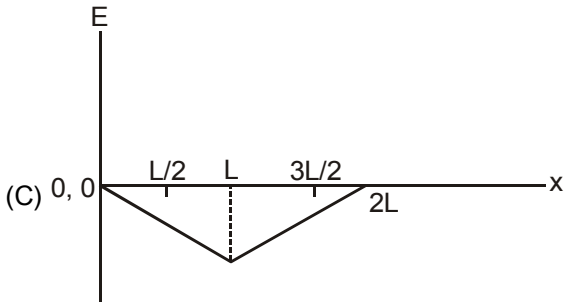
63. A region in the form of an equilateral triangle (in x-y plane) of height L has a uniform magnetic field $\vec{B}$ pointing in the +z-direction. A conducting loop PQR, in the form of an equilateral triangle of the same height L, is placed in the x-y plane with its vertex P at x = 0 in the orientation shown in the figure. At t = 0, the loop starts entering the region of the magnetic field with a uniform velocity $\vec{v}$ along the +x-direction. The plane of the loop and its orientation remain unchanged throughout its motion.

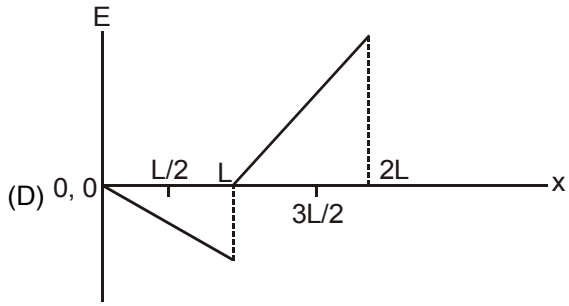


Which of the following graph best depicts the variation of the induced emf (E) in the loop as a function of the distance (x) starting from x = 0. **[JEE-Advanced-2024]**

(A)


(B)


(C)


(D)


64. A coil of area A and N turns is rotating with angular velocity ω in a uniform magnetic field $\vec{B}$ about an axis perpendicular to $\vec{B}$. Magnetic flux φ and induced emf ε across it, at an instant when $\vec{B}$ is parallel to the plane of coil, are : **[JEE-Main-Online-2025]**

(A) $\varphi = AB, \varepsilon = 0$ (B) $\varphi = 0, \varepsilon = NAB\omega$ (C) $\varphi = 0, \varepsilon = 0$ (D) $\varphi = AB, \varepsilon = NAB\omega$

65. Consider I_1 and I_2 are the currents flowing simultaneously in two nearby coils 1 & 2, respectively. If L_1 = self inductance of coil 1, M_{12} = mutual inductance of coil 1 with respect to coil 2, then the value of induced emf in coil 1 will be **[JEE-Main-Online-2025]**

(A) $\varepsilon_1 = -L_1\frac{dI_1}{dt} + M_{12}\frac{dI_2}{dt}$ (B) $\varepsilon_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_1}{dt}$

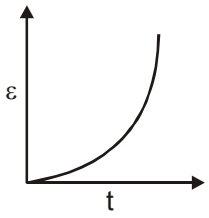
(C) $\varepsilon_1 = -L_1\frac{dI_1}{dt} - M_{12}\frac{dI_2}{dt}$ (D) $\varepsilon_1 = -L_1\frac{dI_2}{dt} - M_{12}\frac{dI_1}{dt}$

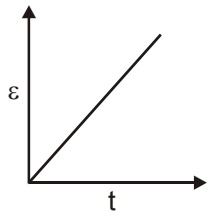
66. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Choke coil is simply a coil having a large inductance but a small resistance. Choke coils are used with fluorescent mercury-tube fittings. If household electric power is directly connected to a mercury tube, the tube will be damaged.

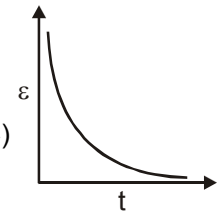
Reason (R) : By using the choke coil, the voltage across the tube is reduced by a factor $\left(\frac{R}{\sqrt{R^2+\omega^2L^2}}\right)$, where ω is frequency of the supply across resistor R and inductor L. If the choke coil were not used, the voltage across the resistor would be the same as the applied voltage.
In the light of the above statements, choose the **most appropriate answer** from the options given below :
(A) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(B) (A) is false but (R) is true. **[JEE-Main-Online-2025]**
(C) Both (A) and (R) are true and (R) is the correct explanation of (A).
(D) (A) is true but (R) is false

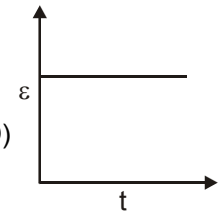
67. A uniform magnetic field of 0.4 T acts perpendicular to a circular copper disc 20 cm in radius. The disc is having a uniform angular velocity of 10 π rad s^{-1} about an axis through its centre and perpendicular to the disc. What is the protential difference developed between the axis of the disc and the rim ? (π = 3.14)
[JEE-Main-Online-2025]
(A) 0.0628 V (B) 0.5024 V (C) 0.2512 V (D) 0.1256 V

68. A rectangular metallic loop is moving out of a uniform magnetic field region to a field free region with a constant speed. When the loop is partially inside the magnate field, the plot of magnitude of induced emf (ε) with time (t) is given by **[JEE-Main-Online-2025]**

(A) 

(B) 

(C) 

(D) 

69. Regarding self-inductance : **[JEE-Main-Online-2025]**
A : The self-inductance of the coil depends on its geometry.
B : Self-inductance does not depend on the permeability of the medium.
C : Self-induced e.m.f. opposes any change in the current in a circuit.
D : Self-inductance is electromagnetic analogue of mass in mechanics.
E : Work needs to be done against self-induced e.m.f. in establishing the current.
Choose the correct answer from the options given below :
(A) A, B, C, D only (B) A, C, D, E only (C) A, B, C, E only (D) B, C, D, E only

70. A solenoid having area A and length 'ℓ' is filled with a material having relative permeability 2. The magnetic energy stored in the solenoid is? **[JEE-Main-Online-2025]**
(A) $\frac{B^2A\ell}{\mu_0}$ (B) $\frac{B^2A\ell}{2\mu_0}$ (C) $B^2A\ell$ (D) $\frac{B^2A\ell}{4\mu_0}$

71. The percentage increase in magnetic field (B) when space within a current carrying solenoid is filled with magnesium (magnetic susceptibility $\chi_{mg} = 1.2 \times 10^{-5}$) is : **[JEE-Main-Online-2025]**
(A) $\frac{6}{5}\times10^{-3}$ % (B) $\frac{5}{6}\times10^{-5}$ % (C) $\frac{5}{6}\times10^{-4}$ % (D) $\frac{5}{3}\times10^{-5}$ %

72. A series LCR circuit is connected to an alternating source of emf E. The current amplitude at resonant frequency is I_0. If the value of resistance R becomes twice of its initial value then amplitude of current at resonance will be : **[JEE-Main-Online-2025]**

(A) I_0 (B) $\frac{I_2}{2}$ (C) $\frac{I_0}{\sqrt{2}}$ (D) $2I_0$

73. An alternating current is given by $I = I_A \sin \omega t + I_B \cos \omega t$. The r.m.s. current will be **[JEE-Main-Online-2025]**

(A) $\sqrt{I_A^2 + I_B^2}$ (B) $\frac{\sqrt{I_A^2 + I_B^2}}{2}$ (C) $\sqrt{\frac{I_A^2 + I_B^2}{2}}$ (D) $\frac{|I_A + I_B|}{\sqrt{2}}$

74. An electric bulb rated as 100 W-220 V is connected to an ac source of rms voltage 220 V. The peak value of current through the bulb is : **[JEE-Main-Online-2025]**

(A) 0.64 A (B) 0.45 A (C) 2.2 A (D) 0.32 A

75. An alternating current is represented by the equation, $i = 100\sqrt{2}\sin(100\pi t)$ ampere. The RMS value of current and the frequency of the given alternating current are **[JEE-Main-Online-2025]**

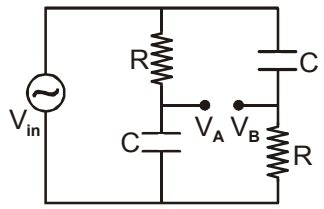
(A) $100\sqrt{2}$A,100Hz (B) $\frac{100}{\sqrt{2}}$A,100Hz (C) 100 A, 50 Hz (D) $50\sqrt{2}$ A,50Hz

76. An ac current is represented as $i = 5\sqrt{2} + 10\cos\left(650\pi t + \frac{\pi}{6}\right)$Amp . The r.m.s value of the current is

(A) 50 Amp (B) 100 Amp (C) 10 Amp (D) $5\sqrt{2}$ Amp

77. An inductor of reactance 100 Ω, a capacitor of reactance 50 Ω, and a resistor of resistance 50 Ω are connected in series with an AC source of 10 V, 50 Hz. Average power dissipated by the circuit is ___ W. **[JEE-Main-Online-2025]**

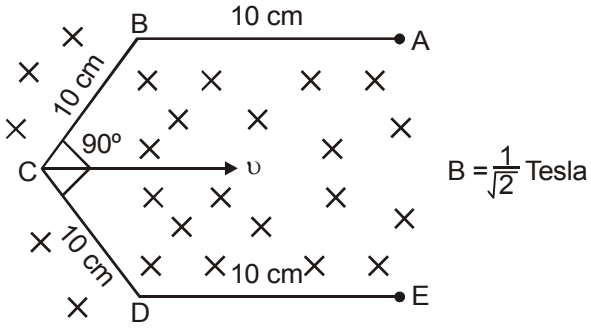
78. For ac circuit shown in figure, R = 100 kΩ and C = 100 pF and the phase difference between V_{in} and $(V_B - V_A)$ is 90°. The input signal frequency is 10^x rad/sec, where 'x' is _____. **[JEE-Main-Online-2025]**



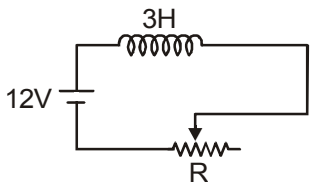
79. In a series LCR circuit, a resistor of 300 Ω, a capacitor of 25 nF and an inductor of 100 mH are used. For maximum current in the circuit, the angular frequency of the ac source is _____ × 10^4 radians s^{-1}. **[JEE-Main-Online-2025]**

80. An inductor of self inductance 1 H connected in series with a resistor of 100 π ohm and an ac supply of 100 π volt, 50 Hz. Maximum current flowing in the circuit is _____ A. **[JEE-Main-Online-2025]**

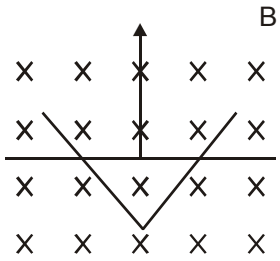
81. Conductor wire ABCDE with each arm 10 cm in length is placed in magnetic field of $\frac{1}{\sqrt{2}}$ Tesla, perpendicular to its plane. When conductor is pulled towards right with constant velocity of 10 cm/s, induced emf between points A and E is _____ mV. **[JEE-Main-Online-2025]**



82. In the given circuit the sliding contact is pulled outwards such that electric current in the circuit changes at the rate of 8 A/s. At an instant when R is 12 Ω, the value of the current in the circuit will be ____ A. **[JEE-Main-Online-2025]**



83. A conducting bar moves on two conducting rails as shown in the figure. A constant magnetic field B exists into the page. The bar starts to move from the vertex at time t = 0 with a constant velocity. If the induced EMF is $E \propto t^n$, then value of n is _____. **[JEE-Main-Online-2025]**

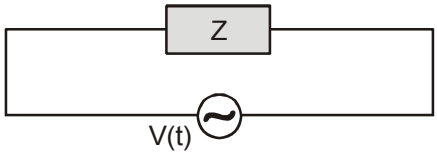


84. The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} Tesla. If the solenoid carries a current of 0.29 A, then the length of the solenoid is _____ π cm. **[JEE-Main-Online-2025]**

85. A tightly wound long solenoid carries a current of 1.5 A. An electron is executing uniform circular motion inside the solenoid with a time period of 75ns. The number of turns per metre in the solenoid is_____. [Take mass of electron $m_e = 9 \times 10^{-31}$ kg, charge of electron $|q_e| = 1.6 \times 10^{-19}$ C, $\mu_0 = 4\pi \times 10^{-7}$ N/A^2, 1 ns = 10^{-9} s] **[JEE-Main-Online-2025]**

86. A circuit with an electrical load having impedance Z is connected with an AC source as shown in the diagram. The source voltage varies in time as V(t) = 300 sin(400t) V, where t is time in s. List-I shows various options for the load. The possible currents i(t) in the circuit as a function of time are given in List-II.

[JEE-Advanced-2025]



Choose the option that describes the correct match between the entries in **List-I** to those in **List-II**

List-I

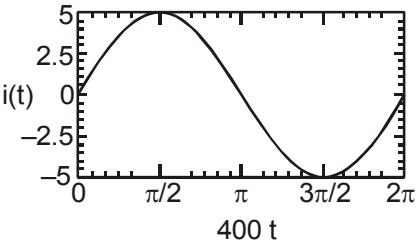
(P) 

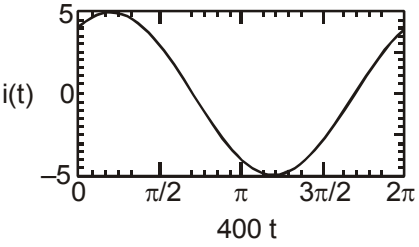
(Q) 

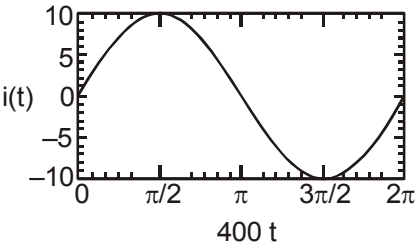
(R) 

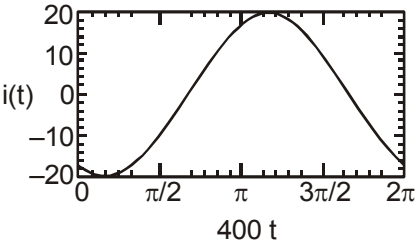
(S) 

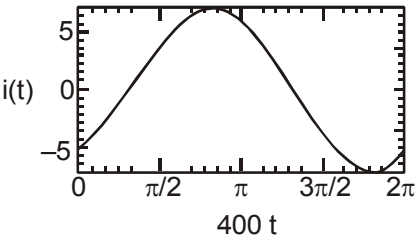
List-II

(1) 

(2) 

(C) 

(D) 

(5) 

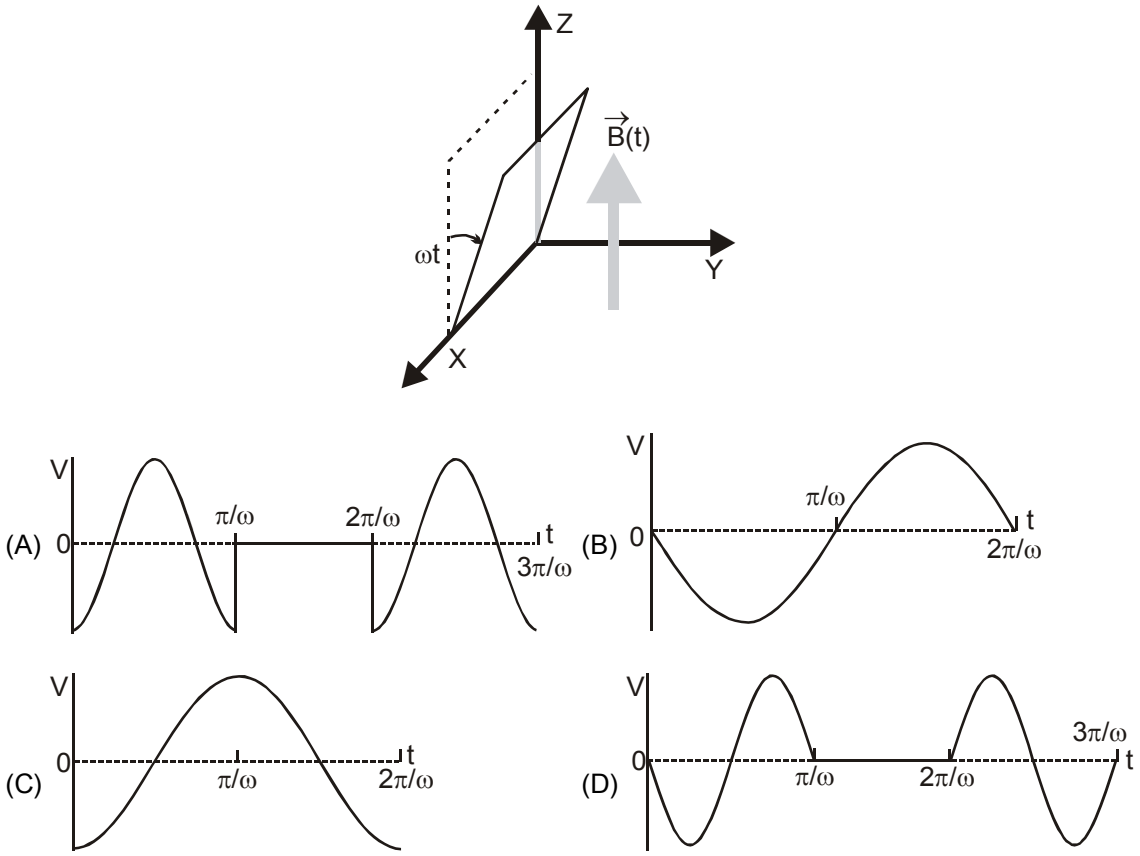
(A) P → 3, Q → 5, R → 2, S → 1

(B) P → 1, Q → 5, R → 2, S → 3

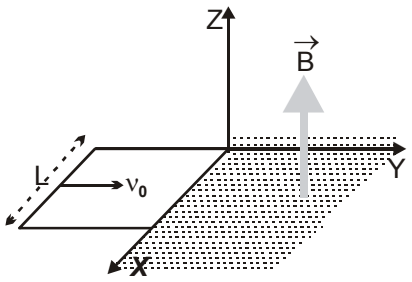
(C) P → 3, Q → 4, R → 2, S → 1

(D) P → 1, Q → 4, R → 2, S → 5

87. A conducting square loop initially lies in the XZ plane with its lower edge hinged along the X-axis. Only in the region $y \geq 0$, there is a time dependent magnetic field pointing along the Z-direction, $\vec{B}(t) = B_0(\cos\omega t)\hat{k}$ where B_0 is a constant. The magnetic field is zero everywhere else. At time t = 0, the loop starts rotating with constant angular speed ω about the X axis in the clockwise direction as viewed from the +X axis (as shown in the figure). Ignoring self-inductance of the loop and gravity, which of the following plots correctly represents the induced e.m.f. (V) in the loop as a function of time : **[JEE-Advanced-2025]**



88. A conducting square loop of side L, mass M and resistance R is moving in the XY plane with its edges parallel to the X and Y axes. The region $y \geq 0$ has a uniform magnetic field, $\vec{B} = B_0\hat{k}$. The magnetic field is zero everywhere else. At time t = 0, the loop starts to enter the magnetic field with an initial velocity $v_0\hat{j}$ m/s, as shown in the figure. Considering the quantity $K = \frac{B_0^2L^2}{RM}$ in appropriate units, ignoring self-inductance of the loop and gravity, which of the following statements is/are correct **[JEE-Advanced-2025]**



(A) If v_0 = 1.5 KL, the loop will stop before it enters completely inside the region of magnetic field.

(B) When the complete loop is inside the region of magnetic field, the net force acting on the loop is zero.

(C) If $v_0 = \frac{KL}{10}$, the loop comes to rest at t = $\left(\frac{1}{K}\right)\ln\left(\frac{5}{2}\right)$.

(D) If v_0 = 3KL, the complete loop enters inside the region of magnetic field at time t = $\left(\frac{1}{K}\right)\ln\left(\frac{3}{2}\right)$.

ANSWER KEY

EXERCISE-I

ONLY ONE OPTION IS CORRECT

1.	C	**2.**	A	**3.**	B	**4.**	A	**5.**	C	**6.**	C	**7.**	C
8.	A	**9.**	A	**10.**	A	**11.**	A	**12.**	B	**13.**	D	**14.**	B
15.	A	**16.**	B	**17.**	A	**18.**	A	**19.**	D	**20.**	B	**21.**	D
22.	B	**23.**	C	**24.**	B	**25.**	C	**26.**	A	**27.**	A	**28.**	A
29.	A	**30.**	B	**31.**	C	**32.**	A	**33.**	B	**34.**	D	**35.**	A
36.	A	**37.**	B	**38.**	D	**39.**	A	**40.**	A	**41.**	A	**42.**	C
43.	D	**44.**	B	**45.**	C	**46.**	A	**47.**	D	**48.**	C	**49.**	B
50.	D	**51.**	C	**52.**	A	**53.**	D	**54.**	B	**55.**	D	**56.**	C
57.	A	**58.**	D	**59.**	D	**60.**	C	**61.**	A	**62.**	A	**63.**	D
64.	C	**65.**	A	**66.**	D	**67.**	C	**68.**	D	**69.**	A	**70.**	A
71.	D	**72.**	C	**73.**	B	**74.**	D	**75.**	D	**76.**	D	**77.**	C
78.	A	**79.**	C	**80.**	C	**81.**	B	**82.**	D	**83.**	B	**84.**	A
85.	A	**86.**	B	**87.**	B	**88.**	C	**89.**	C	**90.**	B	**91.**	B
92.	B	**93.**	D	**94.**	C								

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

95.	B	**96.**	A, C, D	**97.**	A	**98.**	A	**99.**	B, C
100.	B, D	**101.**	A, B	**102.**	B	**103.**	A	**104.**	D
105.	A, B, C, D	**106.**	B, D	**107.**	D	**108.**	A, B, C	**109.**	D
110.	B, D	**111.**	A	**112.**	A	**113.**	A	**114.**	C
115.	B	**116.**	B	**117.**	A, D	**118.**	A, C, D	**119.**	A, C

EXERCISE-II

1. 10 μV **2.** $\lambda V_y B_0$ **3.** 5.0 μV **4.** I^{-1} **5.** 2 N

6. $\frac{\mu_0 i a^2 \pi}{2Rb}$ **7.** $\frac{mgR}{B^2l^2}$ **8.** $\frac{1}{3}$ A **9.** 3μV, clockwise

10. 0.8 **11.** $\frac{LE^2}{2R_1^2}$ **12.** $\frac{1}{15}$ A, $\frac{1}{10}$ A

13. $\frac{erk}{2m}$ directed along tangent to the circle of radius r, whose centre lies on the axis of cylinder.

14. $\frac{e^2}{e^2-1}$ **15.** $\frac{EL}{eR^2}$ **16.** $kMT^2/(R)$ **17.** $LI/2R$

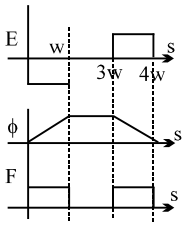
18. (a)10^4A/s (b) 0 (c) 2A (d) $100\sqrt{3}\ \mu$C **19.** $q = Q_0 \sin\left(\sqrt{\frac{1}{LC}}t + \frac{\pi}{2}\right)$ **20.** $\pi/4$

21. $\frac{20}{\pi^2} \cong 2$H **22.** 20 V **23.** R = 100Ω, $\sqrt{3}/\pi$ H

24. C = 9.2 μF

25. (i) 2.4 × 10^{-5} V (ii) from c to b

26.



27. $\frac{\mu_0 h \omega i_m N}{2\pi} \ln \frac{b}{a}$

28. $\frac{\ell}{2}\frac{dB}{dt}\sqrt{R^2 - \frac{\ell^2}{4}}$

29. $\frac{E R_1}{R_1R_2 + R_2R_3 + R_3R_1}$

30. 200 rad/sec

31. (i) $i_1 = i_2 = 10/3$ A, (ii) $i_1 = 50/11$ A ; $i_2 = 30/11$ A, (iii) $i_1 = 0$, $i_2 = 20/11$ A, (iv) $i_1 = i_2 = 0$

32. 42 + 20t volt

33. $I_{EA} = \frac{7}{22}$ A ; $I_{BE} = \frac{3}{11}$A ; $I_{FE} = \frac{1}{22}$A

34. V = 1 ms^{-1}, R_1 = 0.47 Ω, R_2 = 0.30 Ω

35. (a) $E = \frac{1}{2} B\omega r^2$ (b) (i) $I = \frac{B\omega r^2[1 - e^{-Rt/L}]}{2R}$, (ii) $\tau = \frac{mgr}{2}\cos\omega t + \frac{\omega B^2 r^4}{4R}(1 - e^{-Rt/L})$

36. (i) $V_{terminal} = \frac{mgR}{B^2L^2}$; (ii) $\frac{g}{2}$

37. $-\frac{V}{R}e^{-\frac{Rt}{L}}$

38. 0.08 H, 17.28 W

39. 2A, 400W

40. 77Ω, 97.6Ω, 7.7V, 9.76V

41. 0.2 mH, $\frac{1}{32}$ μF, 8 × 10^5 rad/s

42. (a) 10^4 A/s (b) 0 (c) 2A (d) $100\sqrt{3}$ μC

EXERCISE-III

1.	C,D	**2.**	B, C	**3.**	8.00	**4.**	BD	**5.**	BC
6.	CD	**7.**	ABC	**8.**	B,D	**9.**	BCD		
10.	0.63 (0.62 – 0.64)			**11.**	B	**12.**	55.00	**13.**	440
14.	D	**15.**	B	**16.**	C	**17.**	80.00	**18.**	500
19.	74.00	**20.**	3.00	**21.**	B	**22.**	D	**23.**	C
24.	D	**25.**	C	**26.**	A	**27.**	C	**28.**	A
29.	C	**30.**	C	**31.**	C	**32.**	B	**33.**	10.00
34.	C	**35.**	4.00	**36.**	B	**37.**	A	**38.**	08.00
39.	D	**40.**	240	**41.**	B	**42.**	D	**43.**	4.00
44.	D	**45.**	D	**46.**	D	**47.**	0.00	**48.**	4.00
49.	3.00	**50.**	A	**51.**	C	**52.**	A	**53.**	C
54.	A	**55.**	A	**56.**	8.00	**57.**	50.00	**58.**	250.0
59.	10.00	**60.**	22.00	**61.**	10.00	**62.**	A	**63.**	A
64.	B	**65.**	C	**66.**	C	**67.**	C	**68.**	D
69.	B	**70.**	D	**71.**	A	**72.**	B	**73.**	C
74.	A	**75.**	C	**76.**	C	**77.**	1.00	**78.**	05.00
79.	2.00	**80.**	01.00	**81.**	10.00	**82.**	3.00	**83.**	1.00
84.	8.00	**85.**	250	**86.**	A	**87.**	A	**88.**	B, D